A New Method of Solving PDEs

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Abstract

The author has developed an algorithm for finding exact, non-separable solutions of partial differential equations. The algorithm uses Ritt's full reduction algorithm [1] and the GTZ algorithm [2] to find ODEs that solve the PDE. We prove a completeness theorem for the core algorithm, and then give a general form that, using the Rosenfeld–Gröbner algorithm and a terminating recursion, is complete for an arbitrary ansatz. The algorithm has been implemented in software and has been used to find a previously unknown solution to hydrogen's Schrödinger equation.

Introduction

Starting with the following Schrödinger equation for hydrogen [3]:

$$-\frac{1}{2}\nabla^2\Psi - \frac{1}{r}\Psi = E\Psi \tag{1}$$

We can require the solution to solve the following parameterized ODE:

$$(a_0 + a_1 v)\frac{d^2\Psi(v)}{dv^2} + (b_0 + b_1 v)\frac{d\Psi(v)}{dv} + (c_0 + c_1 v)\Psi(v) = 0 \tag{2}$$

where $\Psi(v)$ is a univariate function of some variable v, itself expressed in a parameterized form:

$$v = v_1 x + v_2 y + v_3 z + v_4 r \tag{3}$$

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and expect to find that for certain constant values:

$$\begin{gathered} a_1 = v_4 = 1 \qquad b_0 = c_0 = 2 \qquad c_1 = E \\ a_0 = b_1 = v_1 = v_2 = v_3 = 0 \end{gathered} \tag{4}$$

we will obtain the classical radial equation for hydrogen:

$$r\frac{d^2\Psi(r)}{dr^2} + 2\frac{d\Psi(r)}{dr} + 2(1 + Er)\Psi(r) = 0 \tag{5}$$

a result more commonly derived using separation of variables. Solving the radial equation yields the ground state solution of hydrogen:

$$\begin{gathered} \Psi(r) = e^{-r} \\ E = -1/2 \end{gathered} \tag{6}$$

What might not be expected is to find a different set of constant values:

$$\begin{gathered} E = 0 \\ a_0 = 0 \qquad a_1 = 1 \\ b_0 = -1 \qquad b_1 = 0 \\ c_0 = -1 \qquad c_1 = 0 \\ v_1 = 1 \qquad v_2 = 0 \qquad v_3 = 0 \qquad v_4 = 1 \end{gathered} \tag{7}$$

that produce a different ODE:

$$v\frac{d^2\Psi}{dv}(v) + \frac{d\Psi}{dv}(v) + \Psi(v) = 0 \tag{8}$$

which can be solved to find a previously unknown solution to (1):

$$\Psi = J_0(2\sqrt{x + r}) \tag{9}$$

where x, y, z are Cartesian coordinates, $r = \sqrt{x^2 + y^2 + z^2}$, $E = 0$, and J_0 is the ordinary Bessel function J_0. This solution is not of the familiar spherical separated form $R(r)\Theta(\theta)\Phi(\varphi)$, nor is it Cartesian-separable; it is, however, separable in parabolic coordinates, a reflection of the hydrogen atom's hidden symmetry that we take up in §3.

The author has developed an algorithm that tests a PDE, specified as a differential polynomial, against a system of differential polynomials (which I term an "ansatz"), parameterized by a finite number of constants, similar to (2) and (3), and finds solutions like (4) and (7). This is done using the techniques of differential algebra and algebraic geometry. The restriction to differential polynomials is not severe; most elementary and special functions can be easily modeled, as will be shown below.

In short, to solve the PDE, we restrict our attention to a subset of the function space, parameterized by a finite number of constants and described using differential polynomials. A quick glance at Section 2 should indicate the great variety of such "ansätze". If we can, by luck or by theory, pick a good ansatz, then we have a computable algorithm to find the solutions therein.

Organization of the Paper

Section 1 presents the core algorithm, which fits a given PDE to a given ansatz defined using differential polynomials, and proves its completeness under structural hypotheses on the ansatz; it then gives a general algorithm that discharges those hypotheses for an arbitrary ansatz using the Rosenfeld–Gröbner algorithm and a terminating recursion over the constant space. Section 2 illustrates a half dozen ansätze that can be used. Section 3 illustrates the algorithm by computing a previously unknown solution to the hydrogen Schrödinger equation. Section 4 discusses how the algorithm can be generalized and section 5 discuss the theoretical issues surrounding selection of an ansatz. Section 6 outlines the software used in the example from section 3. Section 7 discusses future work and section 8 is the conclusion.

1 Solution Algorithm

1.1 Differential Algebra Preliminaries

J.F. Ritt introduced differential algebra [1]. Starting with a ring R or a field F, we can construct a *differential ring* or a *differential field* by specifying one or more derivations, unary operators that obey two additional axioms:

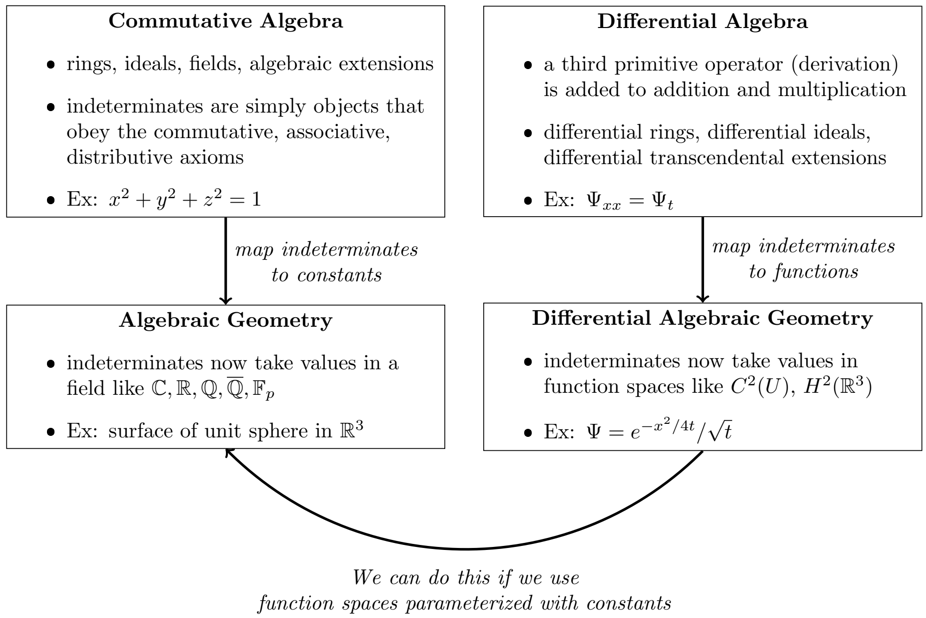

Figure 1: The interrelation between four relevant branches of mathematics

$$dx(a + b) = dx(a) + dx(b)$$
$$dx(ab) = (dx(a))b + a(dx(b))$$

Various notations are in common use for the derivations; since we work mostly in a PDE context where there are multiple derivations, we refer to derivations by prefixing them with a d, i.e, dx. We follow Kolchin ([10] §I.1) and further assume that all derivations are pairwise commutative:

$$dx(dy(a)) = dy(dx(a))$$

Let Θ be the free abelian monoid generated by the derivations; we call the elements of Θ the *derivative operators*, and they act on the differential ring by repeated application of the derivations. The identity element of Θ acts as the identity mapping on the differential ring. The *order* of a differential operator (ord δ) is the number of derivations used to construct it.

A derivative operator applied to an indeterminate is called a *derivative*. We generally use "jet notation" for derivatives, where the applied derivations are denoted by subscripted letters on the indeterminates, i.e, Ψ_x is the result of applying derivation dx to Ψ.

Although we could specify that a derivation maps an indeterminate to some specific element of the differential ring, it more convenient to let each indeterminate introduce an infinite set of derivatives by pairing the indeterminate with all the elements of Θ, so introducing Ψ also introduces Ψ_x, Ψ_y, Ψ_{xy}, Ψ_{xxx}, etc. We can then specify that dx maps Ψ to an element e either by stating that $\Psi_x = e$, or taking the differential ring modulo $\Psi_x - e$.

For each derivation, there will typically exist in the differential ring an associated indeterminate, labeled with the same letter, with the property that applying a derivation to its associated indeterminate produces unity, i.e, $x_x = 1$. Applying a derivation to the associated indeterminate of a different derivation produces zero, i.e, $y_x = 0$. Sometimes it is important to remember that while the same letter is used for both x and dx, they are distinct objects. x is an element of a differential ring, while dx is a unary operator defined on a differential ring.

Analytically, we will always interpret Ψ_x as the derivative of Ψ with respect to x; thinking of the derivation as a unary operator applied to an indeterminate is an algebraic concept. The distinction is much the same as between the real number $\sqrt{2}$ and the element γ in the algebraic extension of $\mathbb{Q}$ defined modulo the ideal $(\gamma^2 - 2)$.

Ritt's convention was to write differential rings by stating their coefficient field, then listing the associated indeterminates of derivations enclosed in curly braces. Each listed indeterminate corresponds to both a derivation (prefixed by d) and its associated indeterminate, i.e, $\mathbb{Q}\{x, y, z\}$ is a differential ring with coefficients in $\mathbb{Q}$, three indeterminates x, y, z and three matching derivations dx, dy, dz. The derivations we often write as subscripts: Ψ_x is the derivation dx applied to Ψ.

1.2 Describing ODEs

Ultimately, we are operating in a partial differential ring, but we are seeking to solve the PDE using ODEs. How, then, do we describe ODEs using differential polynomials? Initially, we consider homogenous linear ODEs, but this is not a fundamental limitation; see below, for example, Figure 9 and equations (30).

An n^{th}-order homogeneous linear ODE takes the form:

$$a_n(v)y^{(n)} + a_{n-1}(v)y^{(n-1)} \cdots a_1(v)y' + a_0(v)y = 0 \tag{10}$$

where $y^{(n)}, y^{(n-1)}, ...y'$ are derivatives of the dependent variable y with respect to the independent variable v. Note that all of the coefficient functions are univariate functions of v. We will further restrict them to be polynomials in v, but this is also not a fundamental limitation; see the ansätze in Figures 7 and 8 and equations (28) and (29).

To describe equation (10) using differential polynomials, we introduce $n+2$ new indeterminates. We need $n+1$ indeterminates for y and its first n derivatives, and we introduce another indeterminate for the *independent variable* v.

We create a differential polynomial in y and its derivatives whose coefficients are polynomials in v with constant coefficients (either numeric constants or constant indeterminates). Thus, the coefficients of the ODE's defining differential polynomial are selected not from the existing field, but rather from $\mathbb{Q}(v)$, an auxiliary field constructed using the independent variable v.

How do the existing derivations operate on the new indeterminates? The independent variable needs no special attention; it is the operation of the derivations on the dependent variable and its derivatives that must be specified. All that required is the chain rule. We need to specify how the new interderminates behave under the existing derivations, and that relationship is analytically:

$$\frac{\mathrm{d}\Psi}{\mathrm{d}x} = \frac{\mathrm{d}\Psi}{\mathrm{d}v}\frac{\mathrm{d}v}{\mathrm{d}x} \tag{11}$$

Algebraically, we write:

$$\Psi_x = \Psi' v_x \tag{12}$$

We use apostrophes, and not the jet notation subscripts, when writing the differential polynomial in the ODE case to emphasize that we are not introducing a new derivation, nor is the ODE's derivative any of the existing derivatives.

In short, to construct a new ODE extension:

- We introduce new indeterminates $\Psi, \Psi', ..., \Psi^{(n)}, v$
- We introduce a new differential polynomial involving only the new indeterminates and constants, and using no other non-constant indeterminates or derivatives
- We define the derivatives of $\Psi, \Psi', ..., \Psi^{(n-1)}$ using the chain rule

Example: In the differential ring $\mathbb{C}\{x, y\}$, we can construct an ODE extension specified by the dependent indeterminate Ψ, its derivative Ψ', independent indeterminate v, and differential polynomial $\Psi' = \Psi$. This differential polynomial maps to the analytic equation $\frac{d\Psi}{dv} = \Psi$ and is solved by the analytic functions Ae^v (A is an arbitrary constant). If $v = x^2 + y$, then the functions Ae^{x^2+y} are solutions to this ODE; the differential polynomials required to model Ae^{x^2+y} are:

$$\Psi_x = 2\Psi' x \qquad \Psi_y = \Psi' \qquad \Psi' = \Psi \tag{13}$$

which, due to the equality of e^v with its own derivative, simplify to:

$$\Psi_x = 2\Psi x \qquad \Psi_y = \Psi \tag{14}$$

1.3 Describing Non-Polynomial Functions

Restricting from differential equations to differential polynomials is not a severe limitation, as so many elementary and special functions can be described as the solutions of differential polynomials. For example, both $y = \cos\theta$ and $y = \sin\theta$ are solutions to the analytic equation $y''(\theta) + y(\theta) = 0$ and can easily be introduced into a system of differential polynomials using the technique of the previous subsection.

For example, consider introducing the function $\cos x$ in the differential ring $\mathbb{Q}\{x, y\}$. We can replace $\cos x$ with a new indeterminate, say f, and write $f'' = -f$, where the derivative is taken with respect to x. The technique from the previous section is used to connect this with the existing derivations, as follows:

$$\begin{array}{cc} f_x = f' & f_y = 0 \\ f'_x = f'' & f'_y = 0 \\ \multicolumn{2}{c}{f'' = -f} \end{array} \tag{15}$$

This can be simplified somewhat by eliminating f'':

$$\begin{array}{cc} f_x = f' & f_y = 0 \\ f'_x = -f & f'_y = 0 \end{array} \tag{16}$$

The analytic solutions to these equations will be of the form $A\cos x + B\sin x$, but this does not present a serious problem. In fact, it is quite common to find a space of solutions for a differential equation, and then to impose additional restrictions (such as boundary conditions) to further restrict those solutions. Likewise, we can use f for our calculations with differential polynomials and, at the conclusion of the calculations, restrict f to be $\cos x$, not to meet a boundary condition, but to force f to be the function required by the original problem.

Likewise, $g = \exp x$ can be expressed using $g' = g$ and $h = \ln x$ using $h' = x'/x$, and the roots of polynomials are solutions to polynomial equations, so most elementary and special functions can be expressed using differential polynomials.

1.4 The Ansatz

The primary intent of the algorithm is find solutions to a PDE in the form of nested extensions, either ODE or algebraic.

We do this by introducing additional differential polynomials into our system, forming what we term the “ansatz”.

Using differential polynomials means that we can't restrict our functions to something like $L^2(\mathbb{R})$, because that requires imposing a global integral condition, and there's no way to specify that using only differential polynomials. Nor can we specify any kind of boundary condition.

We can, however, restrict the function space by requiring the solutions to satisfy additional differential equations.

My primary goal is to “solve” a PDE by finding solutions described using ODEs. Therefore, all of my current ansätze attempt to express solutions to the PDE using a finite combination of ODEs. They do not have to be linear, seperable or homogenous. They do require degree bounds on all of their constituent polynomials, in order to obtain a finite number of constant parameters.

Algebraic extensions can also be included, but since they must be parameterized by a finite number of constants, they can not express an arbitrary algebraic extension of indefinite degree.

To form an ansatz, we construct a tower of ODE and/or algebraic extensions that build from the base differential ring (used to write our original PDE) and culminates in a newly introduced indeterminate that is used as the PDE's dependent variable. It is also possible to consider a system of interrelated PDEs, and multiple parameterized differential polynomials, at the top of the tower, each equated to a new indeterminate and used as the dependent variables in a system of PDEs. (see Figure 9)

The primary requirement for an ansatz is that it be expressed using differential polynomials, as this makes it ammenable to the techniques of differential algebra. This excludes any kind of boundary conditions (which would require evaluating functions at some boundary) or Sobolov space restrictions (which would require evaluating integrals of the function), but admits almost almost any differential equation imaginable.

A good ansatz will be flexible enough to match the solution sought, but require only a small number of parameters, to make the computation tractable.

Several example ansätze will be presented in Section 2.

1.5 Differential Reduction

Having defined an ansatz, we now use it to reduce the PDE. For this purpose, we use Ritt's full reduction algorithm; see [1] Chapter I, §6 or [10], Chapter I, §9. A compact description of the relevant concepts can be found in Section 3 of [11].

Ritt's reduction algorithm requires a *ranking* to be specified on the indeterminates and their derivatives. This is a distinct concept from an ordering on the monomials, as used, for example, in defining a Gröbner basis.

A ranking on a differential ring R is an ordering on the ring's indeterminates and their derivatives that meets two compatibility conditions:

$$\begin{aligned} \forall \delta \in \Theta, \forall u, v \in R \quad & u < v \implies \delta u < \delta v \\ \forall \delta \in \Theta, \forall u \in R \quad & u < \delta u \end{aligned} \tag{17}$$

A ranking is said to be *orderly* if meets the additional condition:

$$\forall \delta, \theta \in \Theta, \forall u, v \in R \quad \text{ord } \delta < \text{ord } \theta \implies \delta u < \theta v \tag{18}$$

Note that an orderly ranking ranks all pure indeterminates lower than any proper derivative.

The *leader* of a polynomial is the highest ranking indeterminate or derivative that appears in the polynomial with a non-zero coefficient.

A differential polynomial p is said to be *partially reduced* with respect to a differential polynomial q if no proper derivative of q's leader appears in p, and *fully reduced* if it is partially reduced and q's leader, if it appears in p, appears at a lower degree than its degree in q.

Ritt's full reduction algorithm reduces a differential polynomial p by a system S of differential polynomials and produces differential polynomials r (the remainder) and h (the "denominator", because it plays that role) such that r is fully reduced with respect to all of the differential polynomials in S and $(hp - r)$ is an element of the differential ideal $[S]$ generated by the elements of S and all their derivatives.

We see that $h = 0$ could lead to extraneous solutions, so we require $h \neq 0$.

1.6 Projection

Having defined an ansatz and used it to reduce the original PDE, we now project the reduced equation onto the space of the n constant parameters $c = (c_1, \ldots, c_n)$. Treating each derivative as a separate indeterminate, we split each term of the remainder r into a coefficient in the constants and a power product in the non-constant indeterminates, then gather the terms that share a power product:

$$r \;=\; \sum_{\alpha=1}^{N} m_\alpha \, p_\alpha(c), \tag{19}$$

where the m_α are the N distinct power products and each coefficient $p_\alpha(c)$ is a polynomial in the constants alone.

Because the m_α are distinct monomials, they are linearly independent over the constants, so r vanishes identically in the non-constant indeterminates if and only if every coefficient does. The projection step therefore replaces the reduced PDE by the system

$$p_1(c) = p_2(c) = \cdots = p_N(c) = 0 \tag{20}$$

of polynomial equations in the constants alone. Satisfying this system makes r vanish; the reduction identity then places h times the PDE in the differential ideal generated by the ansatz, so the ansatz solves the PDE at every choice of constants for which the denominator h does not vanish. Whether, conversely, the algorithm finds *every* solution of the PDE is the question taken up in §1.8.

The system (20) defines an algebraic variety

$$\mathcal{V} \;=\; V(p_1, \ldots, p_N) \;\subseteq\; \mathbb{C}^n,$$

which need not be irreducible. Because only the zero locus matters, we may replace the ideal $(p_1, \ldots, p_N)$ by its radical; the primary decomposition of a radical ideal is a prime decomposition, so its minimal associated primes are genuine primes. Under the well-known correspondence between prime ideals and irreducible varieties ([13], Corollary 4.5.4), these primes cut out the irreducible components of $\mathcal{V}$, and several algorithms have been proposed and implemented to compute them ([2], among others).

1.7 Solution Algorithm

To summarize, the core of the method is the following.

Algorithm 1 (core solution)**.** Given a PDE defined by one or more differential polynomials:

1. Select an ansatz that defines a differential function space parameterized by constants.
2. Reduce the PDE modulo the ansatz (differential elimination step).
3. Construct a system of polynomial equations in the constants by collecting like terms in the remaining variables (projection step).
4. Form a polynomial ideal from that system of equations, and compute its minimal associated prime ideals.
5. Each prime ideal corresponds to an irreducible variety in the space of constants; their union is the solution space of the PDE within the ansatz.

Algorithm 1 runs on any ansatz and always returns a variety in the constants; whether that variety is *complete* — whether it contains every choice of constants for which the ansatz solves the PDE — is the subject of §1.8, guaranteed by Theorem 3 under hypotheses that §§1.9–1.11 show how to discharge for an arbitrary ansatz.

1.8 Completeness of the Projection

A natural question arises: if some particular selection of constants causes the ansatz to solve the PDE, will the algorithm always find those constants in its solution space? Theorem 3 answers it. It covers the regime in which the algorithm of this paper actually operates: every element nonlinear in its leader has a regular initial and a regular separant. Ansätze all of whose leaders are linear — among them the ODE-extension ansätze used here — are always covered, the hypothesis being then vacuous.

We first recall the further notions from differential algebra on which the hypotheses below rest; our terminology and numbering follow [11], §§3–4. The *leader* of a differential polynomial $p \in R \setminus K$ and the notions of a polynomial's being *partially* or *fully reduced* with respect to p were introduced in §1.5. Writing u_p for the leader of p, the *initial* of p is its leading coefficient regarded as a univariate polynomial in u_p, and the *separant* of p is $s_p = \partial p/\partial u_p$. For a set $A \subseteq R \setminus K$ we write I_A, S_A, and $H_A = I_A \cup S_A$ for the sets of initials, separants, and both.

A set $A \subseteq R \setminus K$ is *triangular* if the leaders of its elements are pairwise distinct, and *differentially triangular* ([11], Definition 13) if in addition its elements are pairwise partially reduced.

Two differential polynomials form a *critical pair* ([11], Definition 17) if their leaders have a common derivative. For a critical pair $\{p_1, p_2\}$, write $\theta_1 u$ and $\theta_2 u$ for the two leaders, $\theta_{12} u$ for their least common derivative, and s_1, s_2 for the separants; the associated Δ-*polynomial* ([11], Definition 18) is

$$\Delta(p_1, p_2) \;=\; s_1 \,\frac{\theta_{12}}{\theta_2}\, p_2 \;-\; s_2 \,\frac{\theta_{12}}{\theta_1}\, p_1,$$

where θ_{12}/θ_i denotes the derivation operator carrying $\theta_i u$ to $\theta_{12} u$; its leader is strictly below $\theta_{12} u$. The critical pair is *solved* by a system $A = 0$, $S \neq 0$ ([11], Definition 19) if $\Delta(p_1, p_2) \in (A_v) : (S \cap R_v)^\infty$ for some derivative $v < \theta_{12} u$, where A_v and R_v collect the derivatives of elements of A, respectively all differential polynomials, whose leaders are at most v. A system is *coherent* when every critical pair of A is solved.

Finally, a differential system $A = 0$, $S \neq 0$ is a *regular differential system* ([11], Definition 22) if

(C1) A is differentially triangular;

(C2) S contains the separants of the elements of A and is partially reduced with respect to A; and

(C3) every critical pair of A is solved by $A = 0$, $S \neq 0$ (the coherence property).

The differential ideal $[A] : S^\infty$ is the *regular differential ideal* defined by the system.

Two notions from the theory of (algebraic) triangular sets enter the hypotheses below. Write $\mathrm{sat}(A) = (A) : I_A^\infty$ for the saturation of the algebraic ideal (A) by the initials of A. A triangular set A, with leaders $u_1 < \cdots < u_k$, is a *regular chain* ([14], Definition 5.7) if the initial of each a_i is a non-zero-divisor modulo $\mathrm{sat}(a_1, \ldots, a_{i-1})$, and a *squarefree regular chain* ([14], Definition 7.2) if, in addition, no separant of A is a zero-divisor modulo $\mathrm{sat}(A)$. A regular chain is a characteristic set of $\mathrm{sat}(A)$ — reduction by A is a normal form modulo $\mathrm{sat}(A)$, so a polynomial fully reduced with respect to A and lying in $\mathrm{sat}(A)$ is zero — and a regular chain is squarefree if and only if $\mathrm{sat}(A)$ is radical ([14], Corollary 7.3).

Lemma 2. *Let $A = 0$, $S \neq 0$ be a regular differential system whose coefficients involve constant parameters c, taken over the field of rational functions in those parameters. If c^* is a specialization of the constants at which no initial and no separant of A vanishes, then $A(c^*) = 0$, $S(c^*) \neq 0$ is again a regular differential system.*

Proof. Specialization carries each element of A to its value $A(c^*)$ by evaluating coefficients; it introduces no new derivative, and because no initial vanishes at c^* it leaves every leader in place. Hence $A(c^*)$ remains differentially triangular, and its separants — the specializations of those of A, nonzero by hypothesis — still lie in $S(c^*)$ partially reduced, so conditions (C1) and (C2) persist. For coherence (C3), each Δ-polynomial of A Ritt-reduces to zero modulo A; that reduction is a polynomial identity in the constants whose only multipliers are initials and separants of A ([11], § 3, the separant commuting with the specialization by [11], Lemma 2(e)). Specialized at c^*, where those multipliers do not vanish, the identity shows the Δ-polynomials of $A(c^*)$ likewise reduce to zero. Thus $A(c^*)$ is coherent, hence a regular differential system. □

Lemma 2 is the differential analogue of the "specializes well" property of algebraic regular chains, which holds off the zero set of the iterated resultants of the initials and separants ([20], Definition 11 and Proposition 4; the Gröbner-basis precedent is [21]). Only the differentially-triangular and coherence half is used here; the squarefree regular-chain conditions of hypothesis (4) below need not specialize, and can fail.

Theorem 3. *Let P be a differential polynomial (the PDE), let $A = \{a_1, \ldots, a_k\}$ be the ansatz — a differentially triangular set of differential polynomials whose* *coefficients involve constant parameters $c = (c_1, \ldots, c_n) \in \mathbb{C}^n$ — and let $H_A = I_A \cup S_A$ denote the set of initials and separants of A.*

Apply Ritt's full reduction to P modulo A, obtaining a remainder r and a power product h of elements of H_A such that

$$h \cdot P - r \in [A] \tag{21}$$

with r fully reduced with respect to A. Write $r = \sum_\alpha m_\alpha \cdot p_\alpha(c)$, where the m_α are distinct power products in the non-constant indeterminates and the p_α are polynomials in the constants. Let $\mathcal{V} = V(p_1, \ldots, p_N) \subseteq \mathbb{C}^n$ be the variety defined by setting all constant coefficients to zero.

Suppose $c^* \in \mathbb{C}^n$ satisfies:

1. *$P \in [A(c^*)]$, i.e., the ansatz evaluated at c^* solves the PDE,*
2. *$h(c^*) \neq 0$, i.e., the initials and separants of A do not vanish at c^*,*
3. *A, with its constants regarded as parameters, forms a regular differential system, and*
4. *$A(c^*)$ is a squarefree regular chain.*

Then $c^* \in \mathcal{V}$.

Hypothesis (1) states that the ansatz at c^* "solves the PDE" in the strong sense that P lies in the differential ideal $[A(c^*)]$: every function satisfying the ansatz also satisfies the PDE.

Hypothesis (2) is the "denominator condition" already noted in §1: the initials and separants arising from the Ritt reduction must not vanish. At parameter values where they do vanish, the reduction degenerates and the algorithm may miss solutions.

Hypothesis (3) — regularity — requires the ansatz, with its constants regarded as parameters, to be differentially triangular, with nonzero partially reduced separants, and with all critical pairs solved (the coherence condition; [11], Definition 22, condition C3). We impose it on the parametric ansatz rather than at the single point c^* because, by Lemma 2, hypothesis (2) then carries it to $A(c^*)$. The ODE-extension ansätze considered in this paper satisfy (3) naturally: the chain rule relations and ODE defining polynomial form a differentially triangular system, and the coherence condition is ensured because all cross-derivatives commute consistently through a single independent variable v.

Hypothesis (4) is the ingredient that closes the proof. An element *linear* in its leader satisfies it automatically: its initial and separant coincide, and that quantity is a non-zero-divisor modulo $\operatorname{sat}(A(c^*))$ even when — as can happen — it is a zero-divisor modulo $(A(c^*))$ itself. So (4) bears only on the elements nonlinear in their leaders, of which it asks that *both* the initial and the separant be regular; neither may be dropped, as the two counterexamples following the proof show.

The theorem establishes only the completeness of the algorithm: every valid choice of constants will be found. It says nothing about sufficiency — there may exist points in $\mathcal{V}$ that do not actually solve the PDE, for example because the denominator vanishes there.

Proof. Ritt's full reduction ([1], Chapter I, § 6; [10], Chapter I, § 9; [11], § 3.1) produces h and r satisfying (21), with r fully reduced with respect to A. Evaluating the constants at $c = c^*$ gives $h(c^*) \cdot P - r(c^*) \in [A(c^*)]$. By hypotheses (1) and (2), $h(c^*) \cdot P \in [A(c^*)]$, hence $r(c^*) \in [A(c^*)]$.

Because $r(c^*)$ is fully reduced with respect to $A(c^*)$ it is in particular partially reduced. By hypothesis (2) no initial or separant of A vanishes at c^*, so Lemma 2, applied to the parametric system of hypothesis (3), makes $A(c^*)$ a regular differential system. By Rosenfeld's Lemma ([11], Theorem 23), every partially reduced element of $[A(c^*)] : S^\infty_{A(c^*)}$ lies in the algebraic saturation, so $r(c^*) \in (A(c^*)) : S^\infty_{A(c^*)}$.

By hypothesis (4), $A(c^*)$ is a squarefree regular chain, so its separants are non-zero-divisors modulo $\operatorname{sat}(A(c^*))$; as $(A(c^*)) \subseteq \operatorname{sat}(A(c^*))$ and the latter is therefore unchanged by saturating further at the separants,

$$r(c^*) \;\in\; (A(c^*)) : S^\infty_{A(c^*)} \;\subseteq\; \operatorname{sat}(A(c^*)).$$

But $A(c^*)$ is a characteristic set of $\operatorname{sat}(A(c^*))$, so a polynomial fully reduced with respect to $A(c^*)$ and lying in $\operatorname{sat}(A(c^*))$ must vanish ([14], Theorem 5.13; the vanishing is then immediate from Definition 5.5). Hence $r(c^*) = 0$.

Since the power products m_α are distinct monomials in a polynomial ring (hence linearly independent over $\mathbb{C}$), each coefficient $p_\alpha(c^*) = 0$, whence $c^* \in \mathcal{V}$. $\square$

Regularity of the separants. Hypothesis (4) asks that $A(c^*)$ be *squarefree*, not merely a regular chain. Squarefreeness is what forces the remainder — which Rosenfeld's Lemma places in $(A(c^*)) : S_{A(c^*)}^\infty$ — to lie in $\mathrm{sat}(A(c^*))$, where reduction by $A(c^*)$ recognizes it. Drop it, and the separant saturation can be strictly larger than $\mathrm{sat}(A(c^*))$ and harbour a nonzero fully reduced element.

Counterexample. Take $A = \{u^3 - u^2\}$ in $K[u]$, a single polynomial with leader u of degree 3 and separant $s = 3u^2 - 2u = u(3u-2)$. Then

$$s^2(u-1) \;=\; u^2(3u-2)^2(u-1) \;=\; (u^3-u^2)(3u-2)^2 \;\in\; (A),$$

so $u - 1 \in (A) : S_A^\infty$. The polynomial $u-1$ is fully reduced (its degree in u is $1 < 3$) and nonzero, yet it lies in the saturation: indeed $(A) : S_A^\infty = (u-1)$, strictly larger than $(A) = (u^2(u-1))$. Inverting the separant peels off the factor u^2.

What goes wrong. The separant $s = u(3u-2)$ shares the factor u with A — equivalently, it is a zero-divisor modulo (A), vanishing on the component $u = 0$, an embedded prime of $(A) = (u^2(u-1))$. Saturating by s deletes that component and lowers the degree of u from 3 to 1, leaving the fully reduced $u - 1$.

The remedy. A separant that is a non-zero-divisor cannot do this: it shares no factor with its element, which is then squarefree in its leader, so the leader keeps its full degree under saturation. This is what hypothesis (4) asks of every nonlinear element.

Regularity of the initials. The other half of hypothesis (4) — that $A(c^*)$ be a *regular chain* at all, i.e. that its initials be regular — cannot be dropped either. A triangular set all of whose separants are regular need not be a regular chain; and when it is not, it fails to be a characteristic set of its saturated ideal, so a fully reduced nonzero element can sit in $\mathrm{sat}(A(c^*))$ — indeed in $(A(c^*))$ itself.

Counterexample. In $K[t, w]$ with the ranking $w > t$, take

$$A \;=\; \{\, t^2 - t,\;\; t\,w^2 - w \,\},$$

a triangular set whose leaders are t and w, each of degree 2. Both separants are non-zero-divisors modulo (A): $\mathrm{sep}(t^2 - t) = 2t - 1$ and $\mathrm{sep}(tw^2 - w) = 2tw - 1$ vanish on none of the components listed below. A hypothesis demanding only that the separants be regular would thus be satisfied. Yet

$$w(t-1) \;=\; w^2(t^2 - t) \;-\; (t-1)(tw^2 - w) \;\in\; (A)$$

is fully reduced (of degree $1 < 2$ in each leader) and nonzero. A fully reduced nonzero element therefore sits in (A) — hence in $\mathrm{sat}(A) = (A) : I_A^\infty$ — so A is not a characteristic set of $\mathrm{sat}(A)$ and reduction no longer detects membership.

What goes wrong. The initial of $tw^2 - w$ — the coefficient t of w^2 — is a zero-divisor modulo (A), since $t(t-1) \in (t^2 - t)$. Multiplying $tw^2 - w$ by $t - 1$ annihilates its leading term, $t(t-1)w^2$ being a multiple of the lower generator $t^2 - t$, and the w-degree collapses. The primary decomposition exhibits the loss directly,

$$(A) \;=\; (t, w) \,\cap\, (t-1, w) \,\cap\, (t-1, w-1),$$

three points rather than the $d_t \cdot d_w = 4$ that the reduced monomials would count were A a regular chain: the leaders fail to attain their nominal degrees because the leading coefficient of one element vanishes on a component cut out by another.

The remedy. This is why hypothesis (4) requires $A(c^*)$ to be a *regular chain*, its initials regular and not only its separants. For the ansätze of this paper the requirement holds: the only nonlinear element is the radius relation $r^2 - x^2 - y^2 - z^2$ (initial 1, separant $2r$ a non-zero-divisor), and an element linear in its leader meets both conditions automatically — its initial equals its separant, a non-zero-divisor modulo $\mathrm{sat}(A(c^*))$ even when (as for the ODE separant $a_0 + a_1 v$) it is a zero-divisor modulo $(A(c^*))$.

1.9 Coherence and the Rosenfeld–Gröbner algorithm

Theorem 3 certifies Algorithm 1 under four hypotheses. The first is the premise — the very question we ask of each c^* — and the second, that the reduction denominator not vanish, is the unavoidable price of clearing denominators. The remaining two are structural conditions on the ansatz: that it be a regular differential system (3), and that it specialize to a squarefree regular chain (4). We show in the subsections that follow that neither (3) nor (4) need be imposed on the input. The Rosenfeld–Gröbner algorithm removes (3); a finite recursion over a descending chain of subvarieties of the constant space removes (4). The result is Algorithm 5, which takes an *arbitrary* ansatz and returns, off the denominator locus, exactly the constants at which it solves the PDE.

Hypothesis (3) bundles two requirements: that the ansatz be differentially triangular, and that it be *coherent* (condition C3 of [11], Definition 22). Coherence is the differential analogue of the requirement that every S-polynomial of a Gröbner basis reduce to zero: when two elements have leaders with a common derivative, the two ways of forming that derivative must agree modulo the system. Their difference is the Δ-polynomial, and coherence asks that it reduce to zero.

A failure of coherence is not a contradiction. It means the cross-derivatives expose a constraint not visible in the triangular set itself. The differential analogue of the integrability condition for a gradient field makes this concrete.

Example 4 (the integrability residual)**.** Take $A = \{a_1, a_2\}$ with $a_1 = u_x - f(x, y, u)$ and $a_2 = u_y - g(x, y, u)$ in $K\{u\}$ under the derivations δ_x, δ_y. The leaders $u_x < u_y$ have degree 1 and separant 1, so the set is differentially triangular and pairwise partially reduced. The cross-derivative lies in $[A]$, and full-reducing it modulo A (substituting $u_x \to f$, $u_y \to g$) gives

$$\delta_y a_1 - \delta_x a_2 \;\equiv\; -\Phi \pmod{(A)}, \qquad \Phi \;=\; f_y - g_x + f_u\, g - g_u\, f.$$

Here Φ is the classical integrability residual, the compatibility condition $\partial_y(u_x) = \partial_x(u_y)$ written out. Both initials being 1, the reduction clears no denominator, so $\Phi \in [A]$; and since a_1, a_2 are monic-linear in their leaders, $(A) \cap K[x, y, u] = \{0\}$. The coherence condition for the pair $\{a_1, a_2\}$ reduces (separants being 1) to $\Phi \in (A)$, hence to $\Phi = 0$.

For $f = yu$, $g = 0$ one finds $\Phi = -u \neq 0$: the system is not coherent, $u \in [A]\backslash(A)$ is a hidden algebraic consequence of the cross-derivative, and the system — far from contradictory — is consistent, solved by $u = 0$. When $\Phi = 0$ the system is coherent and no such consequence arises.

A non-coherent ansatz, then, is one whose cross-derivatives imply further equations. These are genuine consequences a faithful analysis must include, yet they are not in general reachable from the triangular set by reduction — which is exactly why hypothesis (3) was imposed. The *Rosenfeld–Gröbner algorithm* [11] removes the need to impose it. Given an arbitrary finite set Σ of differential polynomials, it returns finitely many regular differential systems $(A_1, S_1), \ldots, (A_k, S_k)$ with

$$\{\Sigma\} \;=\; \bigcap_{i=1}^{k} [A_i] : S_i^{\infty},$$

surfacing the hidden constraints as it runs — reducing the Δ-polynomials, splitting when an initial or separant is a zero-divisor — and discarding any inconsistent branch. We may therefore drop hypothesis (3): apply Rosenfeld–Gröbner to the ansatz and treat each resulting component as an ansatz in its own right. For an ansatz that is already a coherent regular differential system the algorithm returns it unchanged, as a single component; the ansätze of §2, built as towers of ODE and algebraic extensions through a single independent variable, are of this kind.

1.10 Specialization and the bad locus

Each component is a regular differential system over the field $\mathbb{C}(c)$ of rational functions in the constants; by the characteristic-decomposition refinement ([15], §7; [14], §§5,7) we may take its triangular set to be a squarefree regular chain there. "There" means *generically* in the constants: squarefreeness is a property of the specialized chain $A(c^*)$, and at special c^* it can lapse, as the *Regularity of the separants* and *Regularity of the initials* counterexamples of §1.8 show. Write

$$B \;\subseteq\; \mathbb{C}^n$$

for the *bad locus*: the constants at which some component fails to specialize to a squarefree regular chain. It is an algebraic variety, cut out by the discriminants of the elements nonlinear in their leaders together with the initials and separants of the *non-monic* elements, eliminated down to the constants by iterated resultants — off the zero set of which a regular chain specializes well ([20]; [21]). An element that is linear in its leader but not monic still contributes here: the locus where its initial vanishes belongs to B even though it carries no discriminant (the ODE element of §3.6 is such a case, giving $B = \{a_0 = a_1 = 0\}$ there). When every nonlinear element has a constant, non-vanishing discriminant and every non-monic initial and separant is a non-zero-divisor, the locus is empty. Off B, hypothesis (4) holds and Theorem 3 applies pointwise.

Recovering the solutions on B is exactly the task of *comprehensive parametric differential elimination*. Restricting the constants to an irreducible component of B and recomputing the characteristic decomposition over its function field returns squarefree regular chains valid at the generic point of that component; because squarefreeness is an open condition, the locus where the recomputed chain in turn degenerates is a *proper* subvariety, of strictly smaller dimension, so iterating terminates after finitely many steps. This recursion — branching on the vanishing and non-vanishing of the initials and separants, the very conditions that cut out B, until every parametric case carries a valid representation — is the *parametric Rosenfeld–Gröbner algorithm* of Fakouri, Rahmany and Basiri [16], and equivalently the *differential Thomas decomposition* of Bächler, Gerdt, Lange-Hegermann and Robertz [17, 18]. Both compute, for an arbitrary parametric differential system, regular representations valid at *every* value of the constants; they are the differential counterpart of the algebraic comprehensive triangular decomposition [20] (in the parametric-Gröbner framework of [19]). We may therefore recover the decomposition on B by invoking either of these algorithms, rather than by an ad hoc recursion.

1.11 The general algorithm

Algorithm 5 (comprehensive solution)**.** Given a PDE P, an ansatz A, and a base field F (initially $\mathbb{C}(c)$):

1. Apply the Rosenfeld–Gröbner algorithm to A over F, obtaining squarefree regular differential chains $C_1, \ldots, C_k$.
2. For each C_i, run Algorithm 1: reduce P modulo C_i, project onto the constants, and decompose, recording the variety of constants at which C_i solves P (valid where the denominator does not vanish).
3. Compute the bad locus B; replace $I(B)$ by its radical and take the irreducible components $B^1, \ldots, B^m$ (the minimal primes).
4. For each B^j, recurse with base field $F := \mathbb{C}(B^j)$, the function field of B^j.
5. Stop when F has transcendence degree 0 over $\mathbb{C}$, or when $B = \emptyset$.

Return the union, over the entire recursion, of the constant sets recorded in step 2.

Steps 1, 3 and 4 are the comprehensive parametric differential elimination of [16, 17]; step 2 is Algorithm 1.

Corollary 6 (completeness for an arbitrary ansatz)**.** *For any ansatz A and PDE P, Algorithm 5 terminates, and the set it returns agrees, off the denominator locus of Theorem 3, with the set of constants c^* at which $A(c^*)$ solves P.*

Proof. By [16, 17] the comprehensive decomposition of steps 1, 3 and 4 terminates and partitions the constant space into finitely many cells, on each of which the chain specializes to a squarefree regular chain. On a cell, where the denominator is nonzero, Theorem 3 places every solving c^* in the set recorded in step 2; conversely, $A(c^*)$ being there a characteristic set of its saturated ideal makes the reduction-to-zero test of Algorithm 1 faithful, so every recorded c^* solves P. The union over cells is thus exactly the solving constants, off the denominator locus. □

Corollary 6 makes the method's completeness unconditional — no hypothesis on the ansatz survives but the denominator condition, which excludes only the degenerate constants where the reduction itself breaks down. (That locus too could be folded into the comprehensive decomposition; we do not pursue it, as it carries no solutions of interest for the ansätze considered here.) We claim nothing about efficiency: Rosenfeld–Gröbner can split, and a recomputation over a function field can be costly. In practice one hopes the ansatz is coherent, so the first step does not split, and that $B = \emptyset$, so no recursion occurs. The hydrogen ansatz of §3 is exactly this case.

1.12 Why not Rosenfeld–Gröbner alone?

The comprehensive decomposition of steps 1, 3 and 4 is a complete differential-elimination engine in its own right, which invites a question: why retain the projection of Algorithm 1 at all? One could instead adjoin the PDE to the ansatz, forming the single differential system $\{P\} \cup A$, and apply comprehensive parametric Rosenfeld–Gröbner to it directly, with the constants c as parameters. Would the solution locus $\mathcal{V}$ not simply be read off the output?

It would — but only after recomputing, by a more expensive route, exactly what the projection computes directly. Reducing P modulo A gives the remainder $r = \sum_\alpha m_\alpha \, p_\alpha(c)$ of (19); over the function field $\mathbb{C}(c)$ each nonzero p_α is a unit, so r enters the chain as a genuine new constraint and the *generic* cell ($c \notin \mathcal{V}$) is the one in which the ansatz does *not* solve the PDE. Tracking the specializations of the constants, the parametric algorithm reaches $\mathcal{V}$ only by branching on $p_\alpha(c) = 0$ in turn; the cell on which *every* coefficient vanishes — where $r \equiv 0$, the PDE becomes redundant, and the chain collapses back to that of the ansatz — is exactly $\mathcal{V}$, and the conditions accumulated along that branch are precisely the projection system (20).

So the two routes agree, off the denominator locus as in Corollary 6; but the comprehensive one obtains the projection equations the hard way, and three things still stand between its output and $\mathcal{V}$. Its native output is an undistinguished forest of parametric cells: singling out the one on which the PDE collapsed, and decomposing it into irreducible solution components, is the projection and prime decomposition of Algorithm 1 recovered as post-processing. That branch is moreover entangled with the bad-locus branching of the ansatz itself (§1.10), two stratifications with no relation to one another. And driving every differential indeterminate down to conditions in the constants forces an elimination ranking, the regime in which Rosenfeld–Gröbner is most prone to splitting and blow-up — whereas the projection is a single Ritt reduction followed by collection of like terms, and in the common case of a coherent ansatz with empty bad locus, which covers every ansatz of §2 and the hydrogen example of §3, the comprehensive machinery is never invoked at all.

The projection, in short, computes directly the one cell that the comprehensive decomposition would reach only by a far more costly detour. The role of the Rosenfeld–Gröbner algorithm in this method is to regularize and certify the ansatz — to supply the squarefree regular chain on which Theorem 3 makes the projection's reduction-to-zero test faithful — not to solve the PDE.

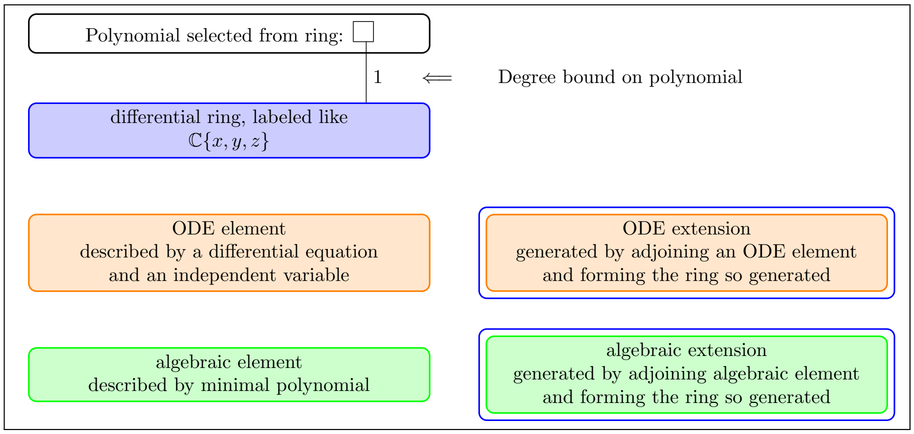

Figure 2: Key to the ansatz diagrams

2 A Collection of Ansätze

There is no one correct ansatz, and currently no theory to guide their selection. Section 5 will discuss some of the difficulties developing such a theory. At the moment, ansatz selection is guided by educated guesses about what function spaces might hold interesting solutions to various PDEs.

Over the next several pages, I'll describe several possible ansätze.

We use a graphical notation to describe an ansatz, for a key see Figure 2. Blue-purple boxes denote rings, orange boxes denote ODE extensions, and green boxes denote algebraic extensions. A blue box drawn immediately around an orange or green box denote the ring formed by adjoining the element defined by the orange or green box. Polynomials are drawn as a square white box and are connected by a line to the ring to which they belong. A black number next to the line indicates a degree bound on the polynomial, and two numbers separated by a slash denote a rational function. For example, 2/2 denotes a rational function with a second degree numerator and a second degree denominator.

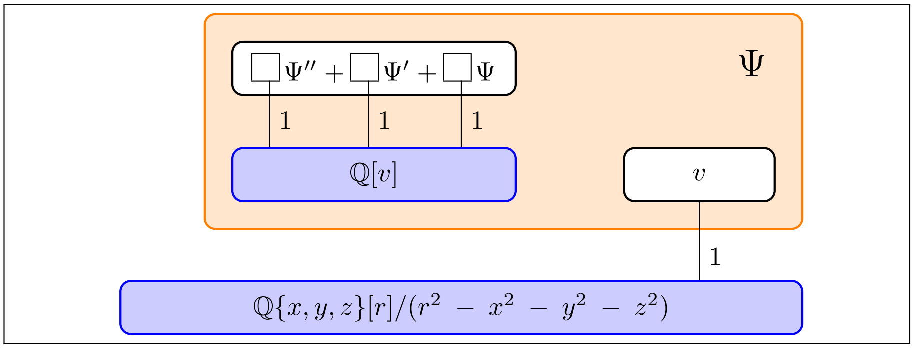

Figure 3: An ansatz for a second-order linear homogenous ODE

The ansatz diagramed in Figure 3 uses a second-order linear homogenous ODE with first degree coefficients and first degree independent variable, and is described using differential polynomials as follows.

First, a differential polynomial involving only Ψ'', Ψ', Ψ, v, and constants, and using no non-constant indeterminates or derivatives.

$$(a_0 + a_1 v)\Psi'' + (b_0 + b_1 v)\Psi' + (c_0 + c_1 v)\Psi = 0 \tag{22}$$

Next, the derivatives of Ψ and Ψ', defined using the chain rule:

$$\begin{array}{ccc} \Psi_x = \Psi' v_x & \Psi_y = \Psi' v_y & \Psi_z = \Psi' v_z \\ \Psi'_x = \Psi'' v_x & \Psi'_y = \Psi'' v_y & \Psi'_z = \Psi'' v_z \end{array} \tag{23}$$

Finally, the independent variable v:

$$v = v_1 x + v_2 y + v_3 z + v_4 r \tag{24}$$

Our differential ring is equipped with three derivations dx, dy, and dz. Our base ring is $\mathbb{Q}\{x, y, z\}[r]/(r^2 - x^2 - y^2 - z^2)$, and from this we select a linear variable v using equation (24). The v_i's (and the a_i's, b_i's and c_i's) are constants. Strictly speaking, a v_0 constant term should also be present, but we often drop the constant term from the independent variable of an ODE extension, because taking the derivative with respect to $x + 3$ is no different from taking the derivative with respect to x.

The element Ψ is expected to solve the PDE, and we use it to write the PDE.

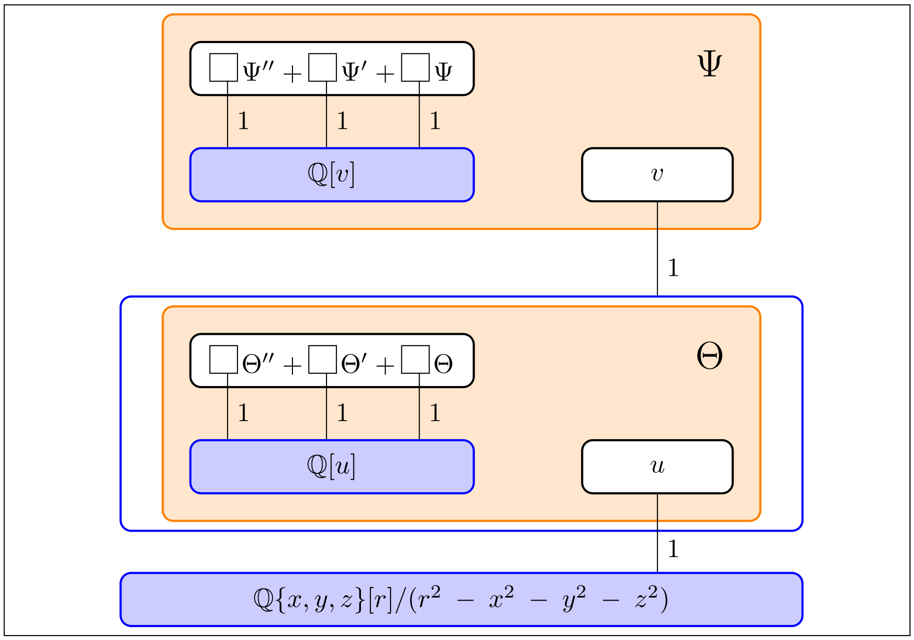

Figure 4: Two nested ODEs

Multiple extensions can be assembled together into a tower. Both ODE and algebraic extensions can be specified. Figure 4's ansatz shows two nested ODE extensions.

$$\Psi_x = \Psi' v_x \qquad \Psi_y = \Psi' v_y \qquad \Psi_z = \Psi' v_z \tag{25a,b,c}$$

$$\Psi'_x = \Psi'' v_x \qquad \Psi'_y = \Psi'' v_y \qquad \Psi'_z = \Psi'' v_z \tag{25d,e,f}$$

$$(a_0 + a_1 v)\Psi'' + (b_0 + b_1 v)\Psi' + (c_0 + c_1 v)\Psi = 0 \tag{25g}$$

$$v = v_1 x + v_2 y + v_3 z + v_4 r + v_5 \Theta \tag{25h}$$

$$\Theta_x = \Theta' v_x \qquad \Theta_y = \Theta' v_y \qquad \Theta_z = \Theta' v_z \tag{25i,j,k}$$

$$\Theta'_x = \Theta'' v_x \qquad \Theta'_y = \Theta'' v_y \qquad \Theta'_z = \Theta'' v_z \tag{25l,m,n}$$

$$(d_0 + d_1 u)\Theta'' + (e_0 + e_1 u)\Theta' + (f_0 + f_1 u)\Theta = 0 \tag{25o}$$

$$u = u_1 x + u_2 y + u_3 z + u_4 r \tag{25p}$$

$$r^2 = x^2 + y^2 + z^2 \tag{25q}$$

Again, we'll write the PDE (not shown) using the expected solution Ψ; that's the convention for the next several examples. We don't write the PDE explicitly because the ansätze can be applied to many PDEs. (30) will shown a different way to construct PDEs.

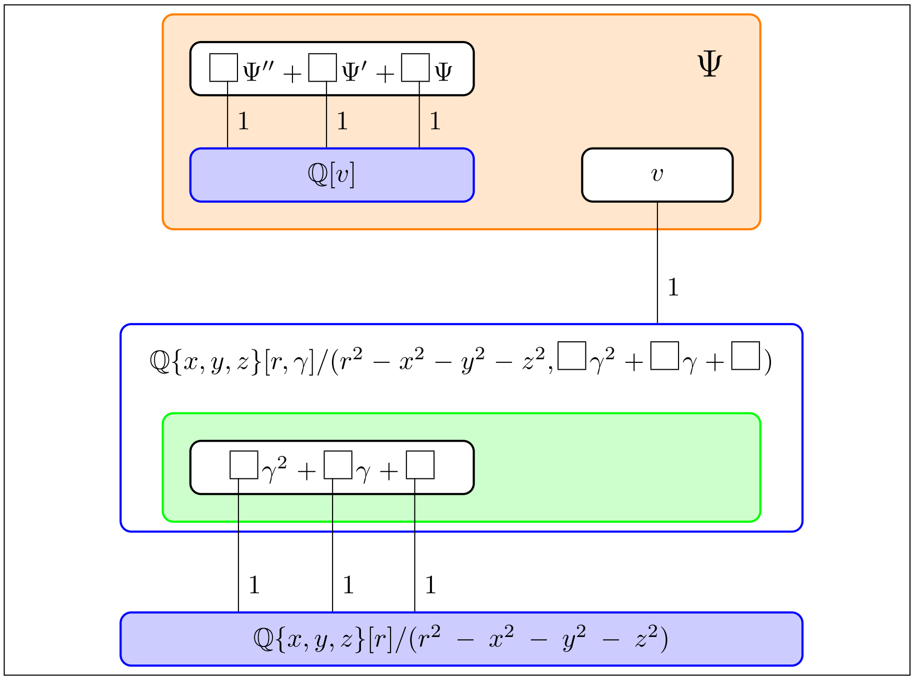

Figure 5: An ansatz for a second-order linear homogenous ODE with first degree coefficients and first degree independent variable, and an algebraic extension used in the formation of the independent variable

Multiple extensions can be assembled together into a tower. Both ODE and algebraic extensions can be specified. Figure 5's ansatz allows a second-degree algebraic extension to be used in the formation of the distinguished variable.

$$\Psi_x = \Psi' v_x \qquad \Psi_y = \Psi' v_y \qquad \Psi_z = \Psi' v_z \tag{26a,b,c}$$
$$\Psi'_x = \Psi'' v_x \qquad \Psi'_y = \Psi'' v_y \qquad \Psi'_z = \Psi'' v_z \tag{26d,e,f}$$
$$(a_0 + a_1 v)\Psi'' + (b_0 + b_1 v)\Psi' + (c_0 + c_1 v)\Psi = 0 \tag{26g}$$
$$v = (v_1 x + v_2 y + v_3 z + v_4 r + v_5 \gamma) \tag{26h}$$
$$g_0 = (g_{00} x + g_{01} y + g_{02} z + g_{03} r) \tag{26i}$$
$$g_1 = (g_{10} x + g_{11} y + g_{12} z + g_{13} r) \tag{26j}$$
$$g_2 = (g_{20} x + g_{21} y + g_{22} z + g_{23} r) \tag{26k}$$
$$g_0 \gamma^2 + g_1 \gamma + g_2 = 0 \tag{26l}$$
$$r^2 - (x^2 + y^2 + z^2) \tag{26m}$$

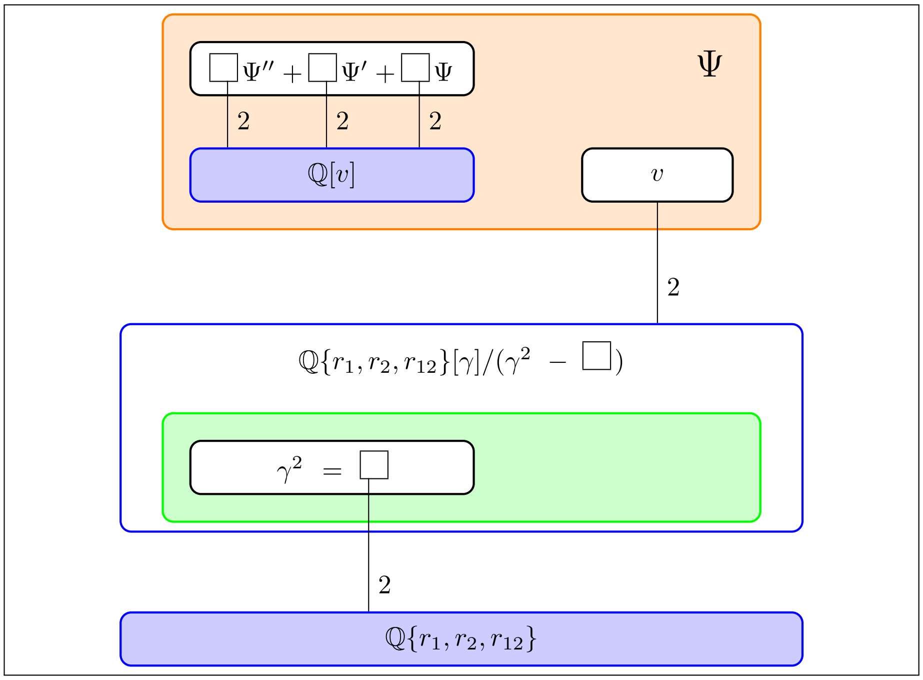

Figure 6: An ansatz for a second-order linear homogenous ODE with second degree coefficients and second degree independent variable, and a square root used in the formation of the independent variable

Second degree algebraic extensions can be specified more concisely using a square root. Higher degree parameterized polynomials can be used, second degree, in this case.

Note that a different set of derivations is used and that no algebraic extension is specified to define r, presumably because it isn't needed to construct the PDE.

$$\Psi_{r_1} = \Psi' v_{r_1} \qquad \Psi_{r_2} = \Psi' v_{r_2} \qquad \Psi_{r_{12}} = \Psi' v_{r_{12}} \tag{27a,b,c}$$

$$\Psi'_{r_1} = \Psi'' v_{r_1} \qquad \Psi'_{r_2} = \Psi'' v_{r_2} \qquad \Psi'_{r_12} = \Psi'' v_{r_{12}} \tag{27d,e,f}$$

$$(a_0 + a_1 v + a_2 v^2)\Psi'' + (b_0 + b_1 v + b_2 v^2)\Psi' + (c_0 + c_1 v + c_2 v^2)\Psi = 0 \tag{27g}$$

$$\begin{gathered} v = v_1 r_1 + v_2 r_2 + v_3 r_{12} + v_4 \gamma + v_5 r_1^2 + v_6 r_2^2 \\ + v_7 r_{12}^2 + v_8 r_1 r_2 + v_9 r_1 r_{12} + v_{10} r_2 r_{12} + v_{11} \gamma r_1 + v_{12} \gamma r_2 + v_{13} \gamma r_{12} \end{gathered} \tag{27h}$$

$$\begin{gathered} \gamma^2 = g_0 + g_1 r_1 + g_2 r_2 + g_3 r_{12} + g_4 r_1^2 + g_5 r_2^2 + g_6 r_{12}^2 \\ + g_7 r_1 r_2 + g_8 r_1 r_{12} + g_9 r_2 r_{12} \end{gathered} \tag{27i}$$

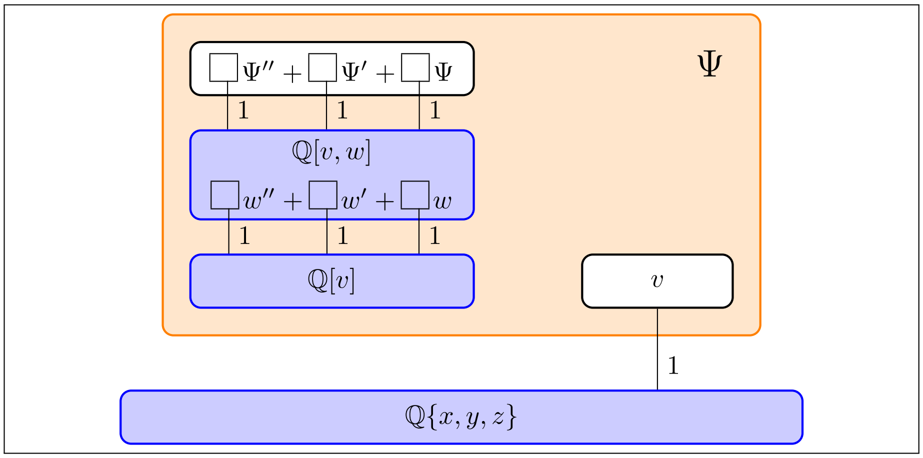

Figure 7: An ansatz with a nested ODE extension

It is also possible to introduce additional extensions into the univariate ring from which the ODE's coefficients are selected (Figure 7), i.e.:

$$
\begin{gathered}
\Psi_x = \Psi' v_x \qquad \Psi_y = \Psi' v_y \qquad \Psi_z = \Psi' v_z \\
\Psi'_x = \Psi'' v_x \qquad \Psi'_y = \Psi'' v_y \qquad \Psi'_z = \Psi'' v_z \\
(a_0 + a_1 v + a_2 w)\Psi'' + (b_0 + b_1 v + b_2 w)\Psi' + (c_0 + c_1 v + c_2 w)\Psi = 0 \\
(d_0 + d_1 v)w'' + (e_0 + e_1 v)w' + (f_0 + f_1 v)w = 0 \\
v = v_1 x + v_2 y + v_3 z
\end{gathered}
\tag{28}
$$

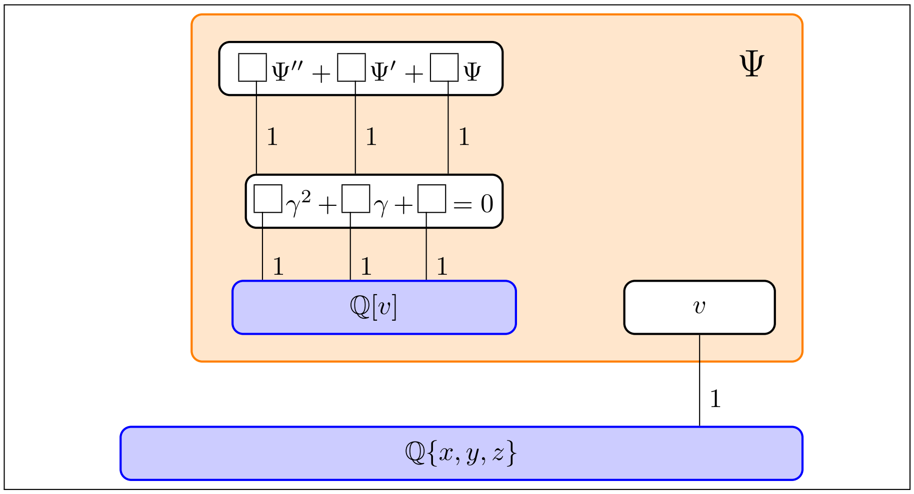

Figure 8: An ansatz with an algebraic extension in the ODE coefficient ring

Algebraic extensions can be introduced in the ODE coefficient ring.

$$\begin{gathered}
\Psi_x = \Psi' v_x \qquad \Psi_y = \Psi' v_y \qquad \Psi_z = \Psi' v_z \\
\Psi'_x = \Psi'' v_x \qquad \Psi'_y = \Psi'' v_y \qquad \Psi'_z = \Psi'' v_z \\
(a_0 + a_1 v + a_2 \gamma)\Psi'' + (b_0 + b_1 v + b_2 \gamma)\Psi' + (c_0 + c_1 v + c_2 \gamma)\Psi = 0 \\
(d_0 + d_1 v)\gamma^2 + (e_0 + e_1 v)\gamma + (f_0 + f_1 v) = 0 \\
v = v_1 x + v_2 y + v_3 z
\end{gathered} \tag{29}$$

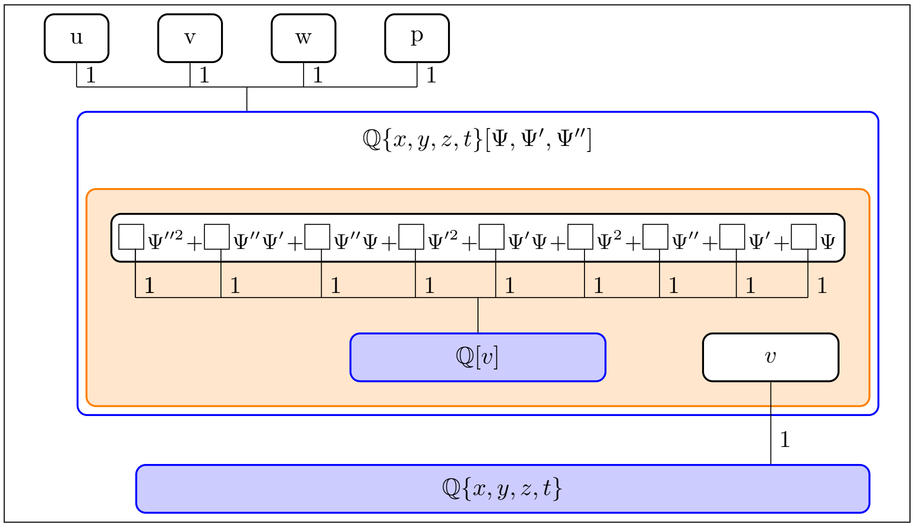

Figure 9: A nonlinear second order second degree homogeneous ansatz

Non-linear ODEs can be constructed. For a linear homogeneous second-order ODE, we have Ψ'', Ψ', and Ψ, multiplied by polynomials in v, so our differential polynomial is first degree in Ψ and its derivatives Ψ' and Ψ''. For a non-linear ODE, we can introduce higher degrees. For example, a second degree second order ODE would add Ψ''^2, $\Psi''\Psi'$, $\Psi''\Psi$, Ψ'^2, $\Psi'\Psi$, and Ψ^2 to the ansatz. See Figure 9.

This time, rather than use Ψ to solve the PDE directly, we define linear polynomials u, v, w, p that will be used to write a system of PDEs.

$$
\begin{gathered}
\Psi_x = \Psi' v_x \qquad \Psi_y = \Psi' v_y \qquad \Psi_z = \Psi' v_z \qquad \Psi_t = \Psi' v_t \\
\Psi'_x = \Psi'' v_x \qquad \Psi'_y = \Psi'' v_y \qquad \Psi'_z = \Psi'' v_z \qquad \Psi'_t = \Psi'' v_t \\
(a_0 + a_1 v)\Psi''^2 + (b_0 + b_1 v)\Psi''\Psi' + (c_0 + c_1 v)\Psi''\Psi \\
+(d_0 + d_1 v)\Psi'^2 + (e_0 + e_1 v)\Psi'\Psi + (f_0 + f_1 v)\Psi^2 \\
+(g_0 + g_1 v)\Psi'' + (h_0 + h_1 v)\Psi' + (i_0 + i_1 v)\Psi = 0 \\
v = v_1 x + v_2 y + v_3 z + v_4 t \\
u = u_1 x + u_2 y + u_3 z + u_4 t + u_5 \Psi \\
v = v_1 x + v_2 y + v_3 z + v_4 t + v_5 \Psi \\
w = w_1 x + w_2 y + w_3 z + w_4 t + w_5 \Psi \\
p = p_1 x + p_2 y + p_3 z + p_4 t + p_5 \Psi
\end{gathered} \tag{30}
$$

3 Example: A New Solution of Hydrogen

We seek to demonstrate the operation of the algorithm on the following Schrödinger equation for hydrogen [3]:

$$-\frac{1}{2}\nabla^2\Psi - \frac{1}{r}\Psi = E\Psi \tag{31}$$

where ∇ is the Laplacian, Ψ is a complex-valued wavefunction defined over real 3-dimensional space, and r is the distance to the origin. We choose to work in Cartesian coordinates, so $r^2 = x^2 + y^2 + z^2$, and we use Hartree atomic units to render the equation dimensionless.

We'll work mostly in the following differential ring:

$$R = \mathbb{Q}\{x, y, z\}[E, a_0, a_1, b_0, b_1, c_0, c_1, v_1, v_2, v_3, v_4, r, \Psi, \Psi', \Psi'', v] \tag{32}$$

To elaborate,

- $\mathbb{Q}\{x, y, z\}$ (and thus R) is a differential ring with derivations dx, dy and dz and matching indeterminates x, y, and z
- $\mathbb{Q}\{x, y, z\}$ is extended by adjoining sixteen indeterminates with no matching derivations
- $E, a_0, .., v_4$ are constants, which are mapped to zero by all derivations
- r, Ψ, Ψ', Ψ'', and v are indeterminates that will ultimately be mapped into a function space, so often we can think of them as functions. All we can really say about them at this point, aside from trivial statements like $dx(\Psi) = \Psi_x$, is that they are not constants.
- Later we will specify a ranking.

3.1 Ansatz

We'll use the first ansatz from the previous section, a second order ODE with first degree coefficients and a first degree independent variable. (Figure 3).

To equations (22), (23) and (24) we append $r^2 - x^2 - y^2 - z^2$ to handle the algebraic extension used to construct the PDE, and obtain the following system of differential polynomials:

$$\Psi_x - \Psi' v_x \qquad \Psi_y - \Psi' v_y \qquad \Psi_z - \Psi' v_z \tag{33a,b,c}$$

$$\Psi'_x - \Psi'' v_x \qquad \Psi'_y - \Psi'' v_y \qquad \Psi'_z - \Psi'' v_z \tag{33d,e,f}$$

$$(a_0 + a_1 v)\Psi'' + (b_0 + b_1 v)\Psi' + (c_0 + c_1 v)\Psi \tag{33g}$$

$$v - (v_1 x + v_2 y + v_3 z + v_4 r) \tag{33h}$$

$$r^2 - (x^2 + y^2 + z^2) \tag{33i}$$

3.2 Ranking

To use Ritt's reduction algorithm, we must now specify a ranking.

We want Ψ'' to be eliminated in favor of Ψ' and Ψ, so we need Ψ'' to be the leader of (33g). This requires that Ψ'' be ranked above Ψ', Ψ, v, and the constants, suggesting

$$\Psi'' > \Psi' > \Psi > v > \text{constants} \tag{34}$$

Also, we want the derivatives of Ψ and Ψ' (like Ψ_x) to be eliminated in favor of Ψ' and Ψ'', so we need Ψ_x to be the leader of $\Psi_x - \Psi' v_x$. This requires that Ψ_x be ranked above both Ψ' and v_x. (34) and (17) already imply that $\Psi_x > v_x$. Ψ_x will be ranked higher than Ψ' if (34) is ranked in an *orderly* manner, meaning that all of the derivatives (like Ψ_x) rank higher than the pure indeterminates (like Ψ and Ψ').

We want the derivatives of v to be eliminated in favor of r, so we need $v > r$. For example, applying dx to (33h), we obtain

$$v - (v_1 x + v_2 y + v_3 z + v_4 r) \xRightarrow{dx} v_x - v_4 r_x \tag{35}$$

v_x will be the leader of (35) if v is ranked greater than both r and the constants, suggesting:

$$\Psi'' > \Psi' > \Psi > v > r > \text{constants} \tag{36}$$

Finally, the `DifferentialAlgebra` library [22] always ranks indeterminates associated with derivations below everything else, so we use this orderly ranking:

$$\Psi'' > \Psi' > \Psi > v > r > \text{constants} > x, y, z \tag{37}$$

3.3 Differential Reduction Step

Applying Ritt's full reduction algorithm to (31) modulo (33) with ranking (37), we obtain the remainder:

$$32(x^2 + y^2 + z^2)p \tag{38}$$

where p is an 85 term differential polynomial that begins:

$$x\Psi' r v_1^3 b_1 + y\Psi' r v_1^2 v_2 b_1 + x\Psi' r v_1 v_2^2 b_1 + y\Psi' r v_2^3 b_1 + \cdots \tag{39}$$

The "denominator" produced by the Ritt algorithm reads:

$$64r^2(va_1 + a_0) \tag{40}$$

so neither r nor $(va_1 + a_0)$ can be zero. In other words, the radius can't be zero, and the ODE's second order coefficient must be non-zero.

3.4 Projection Step

Having reduced PDE (31) modulo (33), we now wish to project our solution into the vector space of the constants. We're looking for constants that will solve equation (39) for all values of x, y, z, r, Ψ, and Ψ', so we collect like terms in x, y, z, r, Ψ, and Ψ', organizing equation (39) like this:

$$r\Psi' x^3(v_1^3 b_1 + v_1 v_2^2 b_1 + v_1 v_3^2 b_1 + 3v_1 v_4^2 b_1) + \cdots \tag{41}$$

The expressions in parenthesis (only one is shown) gives us a system of equations involving only constants that, if satisfied, will yield a solution to (31) in the form (33). Note that E, although not a parameter in the ansatz, is included with them because it is a constant. The correct operation of the algorithm is predicated on nothing more than separating the indeterminates into constants and non-constants, and it makes no difference whether the constants are introduced in the ansatz or appear in the original PDE. The system has 28 equations:

$$
\begin{array}{r}
-2v_1v_4a_1 + 2v_1v_4b_0 \\
-2v_2v_4a_1 + 2v_2v_4b_0 \\
-2v_3v_4a_1 + 2v_3v_4b_0 \\
-2v_4^2a_1 + v_1^2b_0 + v_2^2b_0 + v_3^2b_0 + v_4^2b_0 \\
-2v_4a_0 \\
-2a_0 \\
2v_3v_4c_0 - 2v_3a_1 \\
2v_2v_4c_0 - 2v_2a_1 \\
2v_1v_4c_0 - 2v_1a_1 \\
v_1^2c_0 + v_2^2c_0 + v_3^2c_0 + v_4^2c_0 - 2Ea_0 - 2v_4a_1 \\
4v_2v_3v_4c_1 \\
4v_1v_3v_4c_1 \\
4v_1v_2v_4c_1 \\
v_1^2v_4c_1 + v_2^2v_4c_1 + 3v_3^2v_4c_1 + v_4^3c_1 - 2Ev_4a_1 \\
v_1^2v_4c_1 + 3v_2^2v_4c_1 + v_3^2v_4c_1 + v_4^3c_1 - 2Ev_4a_1 \\
3v_1^2v_4c_1 + v_2^2v_4c_1 + v_3^2v_4c_1 + v_4^3c_1 - 2Ev_4a_1 \\
v_1^2v_3c_1 + v_2^2v_3c_1 + v_3^3c_1 + 3v_3v_4^2c_1 - 2Ev_3a_1 \\
v_1^2v_2c_1 + v_2^3c_1 + v_2v_3^2c_1 + 3v_2v_4^2c_1 - 2Ev_2a_1 \\
v_1^3c_1 + v_1v_2^2c_1 + v_1v_3^2c_1 + 3v_1v_4^2c_1 - 2Ev_1a_1 \\
4v_2v_3v_4b_1 \\
4v_1v_3v_4b_1 \\
4v_1v_2v_4b_1 \\
v_1^2v_4b_1 + v_2^2v_4b_1 + 3v_3^2v_4b_1 + v_4^3b_1 \\
v_1^2v_4b_1 + 3v_2^2v_4b_1 + v_3^2v_4b_1 + v_4^3b_1 \\
3v_1^2v_4b_1 + v_2^2v_4b_1 + v_3^2v_4b_1 + v_4^3b_1 \\
v_1^2v_3b_1 + v_2^2v_3b_1 + v_3^3b_1 + 3v_3v_4^2b_1 \\
v_1^2v_2b_1 + v_2^3b_1 + v_2v_3^2b_1 + 3v_2v_4^2b_1 \\
v_1^3b_1 + v_1v_2^2b_1 + v_1v_3^2b_1 + 3v_1v_4^2b_1
\end{array}
\tag{42}
$$

3.5 Prime Decomposition

We now form an ideal I in $\mathbb{Q}[a_0, ..., v_4, E]$ from (42), and compute its associated prime ideals:

$$(a_0, v_4, v_3, v_2, v_1) \tag{43a}$$

$$(c_1, c_0, b_1, b_0, a_1, a_0) \tag{43b}$$

$$\left(v_1^2 + v_2^2 + v_3^2, a_1, a_0, v_4\right) \tag{43c}$$

$$\left(v_4c_0 - b_0, Ec_0 - v_4c_1, -v_4^2c_1 + Eb_0, b_1, 2a_1 - b_0, a_0, v_3, v_2, v_1\right) \tag{43d}$$

$$\left(v_4c_0 - b_0, v_1^2 + v_2^2 + v_3^2 - v_4^2, c_1, b_1, a_1 - b_0, a_0, E\right) \tag{43e}$$

Several of these varieties solve the system of equations, but do not lead to a meaningful solution to the differential equation. In brief,

(43a) sets the variable v to zero, so we discard it,

(43b) sets all coefficients of the ODE to zero, so we discard it,

(43c) sets the coefficient of Ψ'' in the ODE to zero, resulting in a first-order ODE, so we discard it, too,

(43d) corresponds to the classical solutions (see below), and

(43e) gives us a new solution (see below).

How to understand ideal (43d)? a_0 is an ideal generator, so $a_0 = 0$, and we can always multiply our homogenous differential polynomial by a constant without affecting our result, so we can set $a_1 = 1$. Likewise, we can multiply our variable v by a constant and that will only change our coefficients by constants, and $v_1 = v_2 = v_3 = 0$, so we can normalize by setting $v_4 = 1$. This immediately implies that $b_0 = c_0 = 2$ and simplifies ideal (43d) to:

$$(v_3, v_2, v_1, v_4 - 1, c_0 - 2, 2E - c_1, b_1, 2 - b_0, a_1 - 1, a_0) \tag{44}$$

Setting these polynomials to zero and substituting into (22) and (24), we obtain:

$$\begin{gathered} v = r \\ v\Psi'' + 2\Psi' + 2(1 + Ev)\Psi = 0 \end{gathered} \tag{45}$$

This is the classical radial equation obtained by seperation of variables ([3] §18a). We use spherical coordinates, set $\Psi = R(r)P(\theta)F(\psi)$, and obtain the above equation for $R(r)$, though it is more commonly written in this form:

$$\frac{1}{R}\frac{d}{dr}\left[r^2\frac{dR}{dr}\right] + 2(Er^2 + r) = l(l+1) \tag{46}$$

where l is the orbital quantum number, which can be zero. Set $l = 0$, $v = r$ and $\Psi = R$ and expand out the derivative to obtain (45). This solution has been known for a hundred years; its presence here was expected and won't be commented on further.

Ideal (43e) was not expected, and contains a previously unknown solution. a_0 is an ideal generator, so $a_0 = 0$, and we can set $a_1 = 1$, by the same logic as above.

If we limit our v_i coefficients to be real, all of their squares must be positive, and since $v_1^2 + v_2^2 + v_3^3 - v_4^2 = 0$, v_4 must be non-zero. So we can normalize by setting $v_4 = 1$. Substituting in these values, we find ideal generators that imply $b_0 = 1$ and simplify (43e) to this ideal:

$$\left(v_1^2 + v_2^2 + v_3^2 - 1, v_4 - 1, c_0 - 1, c_1, b_1, 1 - b_0, a_1 - 1, a_0, E\right) \tag{47}$$

This ideal corresponds to the following system of equations:

$$\begin{gathered} E = 0 \\ a_0 = 0 \qquad a_1 = 1 \\ b_0 = -1 \qquad b_1 = 0 \\ c_0 = -1 \qquad c_1 = 0 \\ v_1^2 + v_2^2 + v_3^2 = 1 \qquad v_4 = 1 \end{gathered} \tag{48}$$

Substituting these values back into our ansatz, we conclude that $\Psi(v)$ is a solution of (31) under these conditions:

$$\begin{gathered} v\Psi'' + \Psi' + \Psi = 0 \\ v = v_1 x + v_2 y + v_3 z + r \\ v_1^2 + v_2^2 + v_3^2 = 1 \end{gathered} \tag{49}$$

The expression $v_1x + v_2y + v_3z$ is easily identified as a dot product between the coordinate (x, y, z) and the unit vector (v_1, v_2, v_3) (remembering that $v_1^2 + v_2^2 + v_3^2 = 1$). The direction of this vector is arbitrary, so we can orient the x-axis in this direction and set $(v_1, v_2, v_3) = (1, 0, 0)$ without loss of generality.

We now have to solve a second order ODE:

$$v\Psi''(v) + \Psi'(v) + \Psi(v) = 0 \tag{50}$$

Wolfram Mathematica [23] can now analyze this equation and determine that it is solved by the Bessel function (51).

```
In[1]:= DSolve[x*y''[x] + y'[x] + y[x] == 0, y[x], x]

Out[1]= {{y[x] -> BesselJ[0, 2 Sqrt[x]] C[1] + 2 BesselY[0, 2 Sqrt[x]] C[2]}}
```

leading to the following exact solution to (31), with $E = 0$:

$$\Psi = J_0(2\sqrt{x + r}) \tag{51}$$

where x, y, z are Cartesian coordinates, $r = \sqrt{x^2 + y^2 + z^2}$, $E = 0$, and J_0 is the ordinary Bessel function J_0.

This solution is not of either standard separated form — neither the spherical $R(r)\,\Theta(\theta)\,\Phi(\varphi)$ nor the Cartesian $X(x)Y(y)Z(z)$ — so separation of variables in those coordinates cannot produce it. It is, however, separable in *parabolic* coordinates: with $\xi = r + x$ one of the parabolic coordinates, (51) is a function of ξ alone, the azimuthally symmetric ($m = 0$) separated solution at $E = 0$ ([4], §37). This is no accident, and it is illuminated by the two branches the algorithm returned. The classical branch (43d), with $v = r$, is the spherical separation; the present branch (43e), with $v = v_1 x + v_2 y + v_3 z + v_4 r$, is the parabolic one. That the hydrogen atom separates in *both* systems is the signature of its hidden symmetry: beyond the angular momentum $\mathbf{L}$, the Coulomb problem conserves the Laplace–Runge–Lenz vector $\mathbf{A}$, and together $\mathbf{L}$ and $\mathbf{A}$ generate the dynamical SO(4) symmetry responsible for the degeneracy of the bound spectrum ([5]). Spherical separation diagonalizes $\{L^2, L_z\}$; parabolic separation about an axis $\hat{a}$ diagonalizes instead the components $\mathbf{A} \cdot \hat{a}$ and $\mathbf{L} \cdot \hat{a}$. The algorithm recovers *both* separations from a single ansatz, without committing to any coordinate system in advance — and the unit vector fixing the parabolic axis in the generalized solution (53) is precisely the direction of the Runge–Lenz vector.

The result can be easily verified using Wolfram Mathematica [23], as follows:

```
In[1]:= Psif := Psi[x, y, z]

In[2]:= r[x_, y_, z_] = Sqrt[x^2 + y^2 + z^2]

               2    2    2
Out[2]= Sqrt[x  + y  + z ]

In[3]:= eqn = - 1 / 2 * Laplacian[Psif, {x, y, z}] - 1 / r[x, y, z] * Psif == 0

                                      (0,0,2)                (0,2,0)                (2,0,0)
              Psi[x, y, z]       -Psi       [x, y, z] - Psi       [x, y, z] - Psi       [x, y, z]
Out[3]= -(------------------) + ----------------------------------------------------------------- == 0
                 2    2    2                                  2
          Sqrt[x  + y  + z ]

In[4]:= sol[x_, y_, z_] := BesselJ[0, 2 * Sqrt[x + r[x, y, z]]]

In[5]:= FullSimplify[eqn /. Psi -> ({x, y, z} |-> sol[x, y, z])]

Out[5]= True
```

3.6 Completeness for this ansatz

Having found the solutions above, we close by running the general algorithm of §1.11 on this ansatz and observing that it collapses to a single application of Algorithm 1: the Rosenfeld–Gröbner step returns one component, and although the bad locus is a nonempty codimension-2 subspace, the raw projection covers it directly, so no recursion is needed. The ansatz of Figure 3, given by the differential polynomials (22), (23) and (24), has as its elements a second-order linear homogeneous ODE element

$$(a_0 + a_1 v)\, \Psi'' + (b_0 + b_1 v)\, \Psi' + (c_0 + c_1 v)\, \Psi,$$

with coefficients first-degree in $v = v_1 x + v_2 y + v_3 z + v_4 r$; the chain-rule relations (23) expressing $\Psi_x, \ldots, \Psi'_z$; the variable relation $v = v_1 x + v_2 y + v_3 z + v_4 r$; and the base-ring relation $r^2 - x^2 - y^2 - z^2$ defining the radius r over $\mathbb{Q}\{x, y, z\}$. The two families found solve the PDE in the strong sense $P \in [A(c^*)]$: for the constants of (45) — the classical radial equation, ground state $\Psi = e^{-r}$ — and for those of (43e), the evaluated ansatz reduces the PDE to zero.

The initials and separants. The chain-rule relations (23) and the variable relation (24) are monic and linear in their leaders, so their initials and separants are all 1. Two are nontrivial: the base-ring relation has leader r of degree 2, hence separant $2r$; and the ODE element has leader Ψ'' — ranked above Ψ' and Ψ — of degree 1, so its separant equals its initial, $s = a_0 + a_1 v$.

Rosenfeld–Gröbner returns a single component. Under the orderly ranking (37) the ansatz is differentially triangular: its elements carry the distinct leaders Ψ'', $\Psi'_x, \Psi'_y, \Psi'_z$, Ψ_x, Ψ_y, Ψ_z, v, and r (together with the leaders $v_x, \ldots, r_x, \ldots$ the derivations induce on the lower variables), and the derivative terms in the chain-rule relations contain no proper derivative of any leader. It is coherent: every chain-rule relation factors through the single independent variable v, so the only critical pairs arise from mixed derivatives, which close up because the second derivatives of v are symmetric. The ansatz is therefore already a regular differential system, and the Rosenfeld–Gröbner step of Algorithm 5 returns a single component, without splitting — the regularized (autoreduced) form of the ansatz, with v eliminated through its defining relation and the radius rationalized, rather than, literally, the input system. What matters here is only that there is one component, so Algorithm 1 is applied once.

The bad locus. The only element nonlinear in its leader is the base-ring relation $r^2 - x^2 - y^2 - z^2$. Its initial is 1, and its discriminant in r, $\operatorname{disc}_r(r^2 - x^2 - y^2 - z^2) = 4(x^2 + y^2 + z^2)$, is free of the constants and a non-zero-divisor (the quadric is irreducible, so $\mathbb{Q}[x, y, z, r]/(r^2 - x^2 - y^2 - z^2)$ is a domain), so it contributes nothing to B. The chain-rule and variable relations are *monic*-linear in their leaders (initial 1), and likewise contribute nothing. The ODE element, however, is linear in its leader but *not* monic: its initial equals its separant a_0+a_1v, a nonzero-divisor modulo $\operatorname{sat}(A(c^*))$ at generic c^* — even where it is a zero-divisor modulo $(A(c^*))$ itself — but one that vanishes *identically* when $a_0 = a_1 = 0$, dropping the leader Ψ''. The bad locus is therefore the codimension-2 subspace

$$B \;=\; \{\, a_0 = a_1 = 0 \,\} \;\subseteq\; \mathbb{C}^{11},$$

on which the chain fails to be a squarefree regular chain. Two strata of the decomposition (43) lie in B: (43b) (all ODE coefficients zero) and (43c) (the Ψ'' coefficient zero, the ODE dropping to first order). They are recovered not through hypothesis (4) but because the raw ODE degenerates on B to the *usable* first-order rule $(b_0 + b_1v)\Psi' + (c_0 + c_1v)\Psi$, which the plain Ritt projection of Algorithm 1 follows directly — so the single raw pass already returns the strata on B along with those off it, and no recursive call into B is required.

The denominator. The one residual condition is the denominator of Theorem 3, which the Ritt reduction produced as (40), $64\, r^2(a_0 + a_1v)$: the radius nonzero and the ODE keeping its second-order term, $a_0 + a_1v \neq 0$. Both solutions have $a_0 = 0$, $a_1 = 1$, so $a_0 + a_1v = v$ ($= r$ in the classical case), nonzero away from the origin; both lie off the denominator locus.

Off B, all four hypotheses of Theorem 3 hold and Algorithm 5 reduces to a single application of Algorithm 1; by Corollary 6 the projection is there complete and exact off the denominator locus. On B the chain is no longer a squarefree regular chain, but the same single pass still recovers the degenerate strata (43c) and (43b), the raw ODE dropping to a usable first-order rule. The decomposition (43) is thus complete over the whole constant space, and the two families carrying a genuine second-order ODE — the classical radial equation and the new solution (43e) at $E = 0$ — are all of them.

3.7 Physical Interpretation

Since (51) is a solution to a simple Schrödinger equation for hydrogen, it is reasonable to ask if it has any physical interpretation. (It does not.)

In order to be a physically realistic solution, a function must both solve the Schrödinger equation and be a valid wavefunction. For a given wavefunction Ψ, the modulus squared $|\Psi|^2$ is a probability density function for the position of the electron ([3] §10a), so it follows that a valid wavefunction must be in $L^2(\mathbb{R}^3)$, since its total probability must be finite.

Bessel's original definition of J_0 [24, Eq (10.9.1)] is:

$$J_0(z) = \frac{1}{2\pi} \int_0^{2\pi} \cos(z \sin\theta) d\theta \tag{52}$$

$J_0(z)$ is continuous everywhere on the real line and $J_0(0) = 1$. Also, $2\sqrt{x+r}$ is continuous everywhere in $\mathbb{R}^3$ and is zero along the negative x-axis. Therefore, there is a small tube around the negative x-axis where $J_0(2\sqrt{x+r}) \approx 1$. This tube has positive measure and infinite extent, so $J_0(2\sqrt{x+r}) \notin L^2(\mathbb{R}^3)$, is not valid a wavefunction and does not correspond to any physically realized quantum mechanical state.

The failure to be square-integrable is structural, not accidental. In the parabolic coordinates $\xi = r + x$, $\eta = r - x$ in which (51) separates, our solution is a function of ξ alone; this forces the η separation parameter to vanish, which in turn pins the energy to $E = 0$ ([4], §37), the threshold between the discrete ($E < 0$) and continuous ($E > 0$) spectra. The square-integrable bound states depend on *both* parabolic coordinates — each contributing a decaying confluent hypergeometric factor — and occur only at the discrete energies $E_n = -1/2n^2$. A solution that is a function of a single combination v, and hence of a single parabolic coordinate, therefore cannot be one of them: it necessarily lands on the non-normalizable threshold.

This situation is not unusual. Classical solutions to hydrogen exist (as hypergeometric series) for all energy values, but these solutions are only in $L^2(\mathbb{R}^3)$ for specific values of E; only these wavefunctions are observed as atomic orbitals.

3.8 Generalization of the Example

The choice of x is arbitrary, and any ordinary Bessel function can be used:

$$\Psi = F(2\sqrt{a_1 x + a_2 y + a_3 z + r}) \tag{53}$$

where

$$a_1^2 + a_2^2 + a_3^2 = 1$$

and F is any linear combination of the Bessel functions J_0 and Y_0.

Any finite linear combination of functions of the form (53) also solves (31).

Futhermore, since wavefunctions are complex-valued, restricting the v_i coefficients to be real is unnecessary.

A deeper generalization would enlarge the ansatz itself. The solutions found here are functions of a single combination v, hence of a single parabolic coordinate, and are for that reason confined to the $E = 0$ threshold (§3.7). An ansatz built from *two* ODE-described factors, $\Psi = f_1(v_1)\, f_2(v_2)$ — one for each parabolic coordinate — would instead span the product structure of the square-integrable bound states, whose factors are confluent hypergeometric functions of $\xi = r + x$ and $\eta = r - x$ ([4], §37). Recovering the discrete spectrum $E_n = -1/2n^2$ in this way, with the algorithm discovering the two-coordinate parabolic structure rather than presupposing it, is a natural direction for future work.

4 Generalization of the Algorithm

The algorithm can search for solutions to arbitrary PDEs or systems of PDEs in more complex ansätze. A collection of sample ansätze was presented in Section 2. To handle systems of PDEs, multiple PDEs can be reduced modulo a single ansatz. Each PDE will generate a system of equations in the constants. Combining all of these equations together will produce a system of equations that, when satisfied, will solve all of the PDEs simultaneously.

Multiple ODE extensions can be introduced, either in parallel, allowing the ODE solutions to be combined together into a polynomial that solves the PDE, or in series (Figure 4), allowing either the independent variable of one ODE to be formed from the solutions to a second ODE. Extensions can also be introduced in the coefficient ring of the ODE (Figures 7 and 8)

In addition to ODE extensions, algebraic extensions can be introduced either explicitly or implicitly. Figure 5 shows an explicit second degree algebraic extension. Introducing higher degree algebraic extensions is also possible. Figure 6, a somewhat simpler variant on the same idea, shows an ansatz that introduces a square root, then allows the ODE's independent variable to be selected from the resulting algebraic extension.

Parameterized polynomials of any (finite) degree can be used in lieu of the linear polynomials. Figure 6 illustrates the use of second degree polynomials. Polynomials can be further restricted. Perhaps we have some theoretical consideration which suggests that the independent variable should be symmetric.

Nonlinear ODEs can be introduced, as shown in Figure 7.

Nonlinear ODEs permit algebraic extensions to be introduced implicitly. Consider that for a given algebraic extension:

$$f_n(x)y^n + \cdots + f_1(x)y + f_0(x) \qquad f_i(x) \in \mathbb{C}[x] \tag{54}$$

we can form the derivative with respect to x as follows:

$$y' = \frac{f_n'(x)y^n + \cdots + f_1'(x)y + f_0'(x)}{nf_n(x)y^{n-1} + \cdots + f_1(x)} \tag{55}$$

This is a non-linear ODE that is first order, n^{th} degree in y.

Note that if we limit our ansatz to linear ODEs, then any algebraic extensions need to be introduced explicitly; i.e, a linear ODE will not match algebraic extensions.

We can introduce inhomogenous ODEs very similarly to the nonlinear case by introducing a zeroth degree coefficient into the differential equation.

Do we need to restrict the coefficients of the ODE to a ring generated by the independent variable? We could allow coefficients in the ODE to involve indeterminates other than v, and there are two major cases to consider. The first case is when we can partition the indeterminates into two sets, one used to express the independent variable, and one used to express the coefficients of the ODE. This yields a *parameterized ODE*, one whose coefficients vary according to one set of independent variables (the parameters), and whose independent variable depended on a different set of them. The parameters can be treated as constants for the purpose of solving the ODE. I don't know of any way to enforce this requirement other than having multiple ansätze, each with a different partitioning of the indeterminates. One large ansatz could be constructed that would enforce nothing, but would still allow for ODE solutions of the form described next.

What if the dependencies of the ODE coefficients and the ODE variable can't be partitioned into two different sets of variables?

Then we're going to get an ODE something like this (two dimensional, for illustrative purposes):

$$f(u,v)\frac{d^2\Psi}{dv^2} + g(u,v)\frac{d\Psi}{dv} + h(u,v)\Psi = 0 \tag{56}$$

the point being that we can't separate the v dependency out of the coefficients.

I'm not aware of any theory to handle an ODE of the form (56), but one might exist, or be developed. Would it be useful? Perhaps. (56) is still simpler than a PDE because it only involves one derivative. The algorithm described in this paper could test for the existence of such solutions. I'm somewhat skeptical of their utility.

5 Selection of an Ansatz

The most obvious question remains: how to select an ansatz?

First, can we build any such ansatz to express all of the solutions of the PDE? This seems unlikely, as the solutions of a PDE would include solutions for all possible boundary conditions.

For example, the simple heat equation $u_t = u_{xx}$ admits an infinity of solutions of the form

$$u(x,t) = \frac{1}{\sqrt{4\pi t}} e^{-\frac{(x-x_0)^2}{4t}} \qquad \forall x_0 \in \mathbb{R} \tag{57}$$

No single exponential extension can be constructed to express all of these solutions. In fact, a different extension is required for each real value of x_0, yielding a uncountable infinite basis.

Furthermore, these are just the solutions described in a series expansion; we expect there to be many others. The essence of the problem of ansatz selection is that we expect there to be solutions all over the place. Many ansätze will yield solutions.

So, while the PDE problem is superficially similar to the problem of symbolic integration, which asks if a finite tower of extensions can be used to express an integral, the nature of the problem is quite different because in the integration case we can express all of its solutions in a single extension. In fact, given one solution to the integration problem, we know that all of its solutions are related by adding a constant C, which implies that a field containing one solution contains all solutions. In the symbolic integration case, we expect to find a single field extension that contains all solutions, while in the PDE case we expect this to be impossible.

What about differential Galois theory?

Picard-Vessiot / differential Galois theory has been of great utility in analyzing ordinary differential equations. An n^{th} order linear ODE, will have at most an n-dimensional solution space. After selecting an arbitrary basis, any solution can be represented by an n-tuple of complex constants. Any automorphism that permutes these solutions can be represented by an $n \times n$ matrix, so differential Galois theory can be reduced to the analysis of $GL_n(\mathbb{C})$, which has facilitated the work of Kovacic, Singer, and Bronstein.

No such easy reduction is available for PDEs. A PDE with no boundary conditions is likely to have a solution space that is not only infinite, but infinite dimensional, and probably not even with a countably infinite basis. In such an environment, differential Galois groups can not be represented as subgroups of GL_n, so the theory of algebraic groups is therefore unlikely to be much help.

Kovacic developed an algorithm [25] to handle second order ODEs not by developing an algorithm to determine the differential Galois group of a given equation, but rather by using first a change of variables to restrict the Galois group to SL_2, then categorizing all possible algebraic subgroups of SL_2 and developing an algorithm to handle each case. Taking a similar approach for PDEs would require categorizing all possible linear mappings between function spaces, which does not seem feasible.

In short, the differential algebra techniques used to analyze integrals and linear ODEs do not seem amenable to handling PDEs, and no theory is available at this time to guide the selection of an ansatz. The ansatz used for the hydrogen example was an educated guess. It was designed to be general enough to find the existing classical solution, and so to be used for testing the software. The appearance of (43e) was not anticipated, as the solutions was previously unknown, and serves to illustrate the mysterious nature of these ansätze.

6 Software

The script used to perform the calculations in this paper is available here:

`https://github.com/BrentBaccala/helium`

`joca.sage` is a SageMath script, verified to work with SageMath 10.6.

SageMath [26] integrates dozens of open source mathematical programs and libraries into a unified, Python-based framework. For this calculation, it uses François Boulier's `DifferentialAlgebra` package [22] to perform the Ritt reduction step, and Singular [27]'s implementation of the GTZ algorithm [2] to compute the associated prime ideals of (42)

7 Future Work

The difficulties of selecting an ansatz have already be touched upon. One possible approach to ansatz selection would be to impose boundary conditions (or the like) prior to any kind of analysis with differential algebra. In the case of quantum mechanics, imposing $H^2(\Omega)$, where Ω is the state space of whatever physical system we're analyzing, is an obvious restriction. Some kind of theoretical analysis, exploring what restrictions must be placed on ansätze to force the functions into $H^2(\Omega)$, might yield useful techniques to better select ansätze.

What relationship is there between the constant ideals and the function space?

The ranking was selected in somewhat of an ad hoc manner. A more systematic method of analyzing an ansatz to determine a suitable ranking would be useful.

The algorithm, insofar as it relies on separating constant and non-constant indeterminates, produces correct solutions, but there is no proof that all possible solutions will be found. Different rankings produce systems with more non-constant indeterminates, suggesting that finding all possible solutions places requirements on the ranking.

While discarding ideals (43a) and (43b) seems justified (the independent variable must not be zero, and the ODE must have some non-zero coefficients), discarding ideal (43c) (setting the ODE's second-order term to zero) seems somewhat arbitrary. We would like to analyze the first-order ODE case as a special case of the second-order ODE. The Rosenfeld-Gröbner algorithm [11] may provide a suitable tool for such an analysis, as it decomposes a system of differential polynomials into a family of overlapping systems with regularity properties that include requiring the *initials* of the differential polynomials to be non-zero. For an n-order ODE, requiring its initial to be non-zero would require its n-th order term to be non-zero, so Rosenfeld-Gröbner should split (43c) into a separate system. Unfortunately, our current implementation of Rosenfeld-Gröbner ([22]) runs for five days on this system without termination.

The utility of this algorithm is currently limited by our Gröbner basis software. For example, I have formed a system of 1570 equations with 1,296,960 terms to test whether helium can be solved using ansatz (27). Various attempts to compute a decomposition of this system into its minimal associated primes exhaust the memory on a computer with 96 GB RAM, but the software lacks the ability to fall back effectively on mass storage when RAM is exhausted, nor can it use memory spread out over a cluster.

8 Conclusion

The algorithm presented above can be used to check any PDE to see if any of its solutions can be expressed using an ODE structured according to a specific ansatz, defined by differential polynomials and parameterized by a finite number of constants. This technique is complementary to separation of variables, where we check a PDE to see if any of its solutions can be expressed as a product of factors, each depending on only a single variable, with no restrictions to differential polynomials. It is likewise complementary to the *similarity-reduction* methods — the classical Lie symmetry method [6], the Clarkson–Kruskal direct method [7, 8], and the nonclassical (conditional) symmetries of Bluman and Cole [9] — which also reduce a PDE to an ODE without assuming a separable product form, but which are organized around the (classical or conditional) symmetries of the equation. Our reductions are instead organized around the differential-algebraic structure of the ansatz; they do not presuppose any symmetry of the PDE, and they carry the completeness guarantee of Theorem 3.

The name of the software repository (`helium`) informs a primary goal of my research: to find a closed form expression for helium's ground state. In that repository, you will find several years of attempts in that direction, of which `joca.sage` is the most refined. Although "solving helium" is widely regarded as impossible, I know of no proof that helium's ground state wavefunction can not be formed from a finite number of linear homogenous ODEs. Lacking any good theory to inform the selection of an ansatz, I have turned to educated guesswork.

We can explore other important PDEs, such as the Navier-Stokes equation. As with helium, we can work incrementally towards it using smaller ansätze. Ansatz (30) is a step in the direction of analyzing Navier-Stokes. $\bar{\mathbf{v}} = (u, v, w)$ is meant to be a velocity field and p is the pressure, all varying as functions of position (x, y, z) and time t.

We can solve differential equations with Gröbner bases, and this technique, given enough compute resources, allows us to check for simple ODE-based solutions, but we're primarily limited by inefficiencies of our software.

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A Source Code

```
# -*- mode: python -*-
#
# Sage script to do the computations for my paper in the
   Journal of Computational Algebra.
#
# Author: Brent Baccala
# Date: December 12, 2025
#
# Tested on Ubuntu 24 with Sage 10.6
#
# François Boulier's Differential Algebra package is
   required to run this script.
# To install it, run this command from the sage prompt:
#
# %pip install DifferentialAlgebra
#
# DifferentialAlgebra (sympy based) is used to perform the
   differential algebra reduction.
# Sage is used to compute a primary decomposition of the
   resulting system of equations.
# Some fiddling is required to juggle back and forth between
    the two.

import sympy

try:
    import DifferentialAlgebra
except ModuleNotFoundError as ex:
    raise ModuleNotFoundError(ex.msg + "\nInstall␣it␣with␣'%
        pip␣install␣DifferentialAlgebra'")

# Claude Sonnet 4's solution to printing "DPsi" as "\Psi'"
   in LaTeX

def patch_latex_varify():
    """
    Patch sage.misc.latex.latex_varify to handle 'DPsi'
        specially.
    """
    from sage.misc.latex import latex_varify
    import sage.misc.latex

    # Save reference to the original function
    original_latex_varify = latex_varify

    # Define the custom function
    def custom_latex_varify(a, is_fname=False):
```

```
        if a == "DPsi":
            return r"\Psi'"
        else:
            return original_latex_varify(a, is_fname=
                is_fname)

    # Replace the function in the module
    sage.misc.latex.latex_varify = custom_latex_varify

patch_latex_varify()

# Declare our independent variables
x,y,z = sympy.var('x,y,z')

# Declare our constants
E = sympy.var('E')
v1,v2,v3,v4 = sympy.var('v1,v2,v3,v4')
a0,a1,b0,b1,c0,c1 = sympy.var('a0,a1,b0,b1,c0,c1')
constants = [E,v1,v2,v3,v4,a0,a1,b0,b1,c0,c1]

# Declare our dependent variables
#
# `indexedbase` is DifferentialAlgebra's suggested way of
    declaring dependent variables
# if we want to use jet notation (i.e, v[x] for dv/dx) to
    write their derivatives.

Psi,DPsi,DDPsi = DifferentialAlgebra.indexedbase('Psi,DPsi,
    DDPsi')
v = DifferentialAlgebra.indexedbase('v')
r = DifferentialAlgebra.indexedbase('r')

# Create the DifferentialRing and set the ranking on the
    variables

DiffRing = DifferentialAlgebra.DifferentialRing (derivations
     = [x,y,z],
                                                  blocks = [[
                                                      DDPsi,
                                                      DPsi,Psi
                                                      ,v,r],
                                                      constants
                                                      ],
                                                  parameters
                                                      =
                                                      constants
                                                      ,
                                                  notation =
                                                      'jet')
```

```
# Define the PDE we're trying to solve
# sympy can't handle a Sage Integer, so use casts to make
    these Python integers

PDE = -int(1)/int(2)*(Psi[x,x] + Psi[y,y] + Psi[z,z])*r -
    Psi - E*r*Psi

print("PDE:", PDE)

# Define the ansatz, the parameterized function space in
    which we're looking for solutions

ansatz = [Psi[x] - DPsi * v[x],
          Psi[y] - DPsi * v[y],
          Psi[z] - DPsi * v[z],
          DPsi[x] - DDPsi * v[x],
          DPsi[y] - DDPsi * v[y],
          DPsi[z] - DDPsi * v[z],
          (a0 + a1*v)*DDPsi + (b0 + b1*v)*DPsi + (c0 + c1*v)
              *Psi,
          v - (v1*x + v2*y + v3*z + v4*r),
          r**2 - x**2 - y**2 - z**2]

# DifferentialAlgebra can't handle the parenthesized
    expressions directly, so expand them

ansatz = list(map(sympy.expand, ansatz))

print("\nAnsatz:", *ansatz, sep='\n')

# Reduce the PDE modulo the ansatz using Ritt's reduction
    algorithm

h,r = DiffRing.differential_prem(PDE, ansatz)

print("\nRemainder:", r)

# Convert the remainder to Sage

PolyRing = PolynomialRing(QQ, names=[str(indet) for indet in
     DiffRing.indets(selection='all')])
PolyRing_constants = list(map(PolyRing, constants))
PolyRing_r = PolyRing(r)

# Given an equation and a list of constants, factor each
    term into constant and non-constant factors,
# then group together all terms with identical non-constant
    factors and return the resulting
# list of equations (which will only involve constants).
```

```
def build_system_of_equations(eqn, constants):
    ring = eqn.parent()
    system_of_like_terms = dict()
    non_constant_sub = tuple(1 if ring.gen(n) in constants
        else ring.gen(n) for n in range(ring.ngens()))
    for coeff, monomial in eqn:
        non_constant_part = monomial(non_constant_sub)
        constant_part = coeff * monomial //
            non_constant_part
        if (non_constant_part) in system_of_like_terms:
            system_of_like_terms[non_constant_part] +=
                constant_part
        else:
            system_of_like_terms[non_constant_part] =
                constant_part
    return tuple(set(system_of_like_terms.values()))

eqns = build_system_of_equations(PolyRing_r,
    PolyRing_constants)

print("\nSystem␣of␣equations:", *eqns, sep='\n')

# Build a polynomial ideal from the system of equations and
    construct its prime decomposition

I = ideal(eqns)
prime_decomposition = I.minimal_associated_primes()

# Sort this result (so it prints in the same order as in the
     JOCA paper) and print it

prime_decomposition.sort(key=lambda x:str(x))
print("\nMinimal␣associated␣prime␣ideals:", *
    prime_decomposition, sep='\n')
```